INSTITUTO DE FOMENTO PESQUERO

2nd International Symposium on Squid Conservation and Management

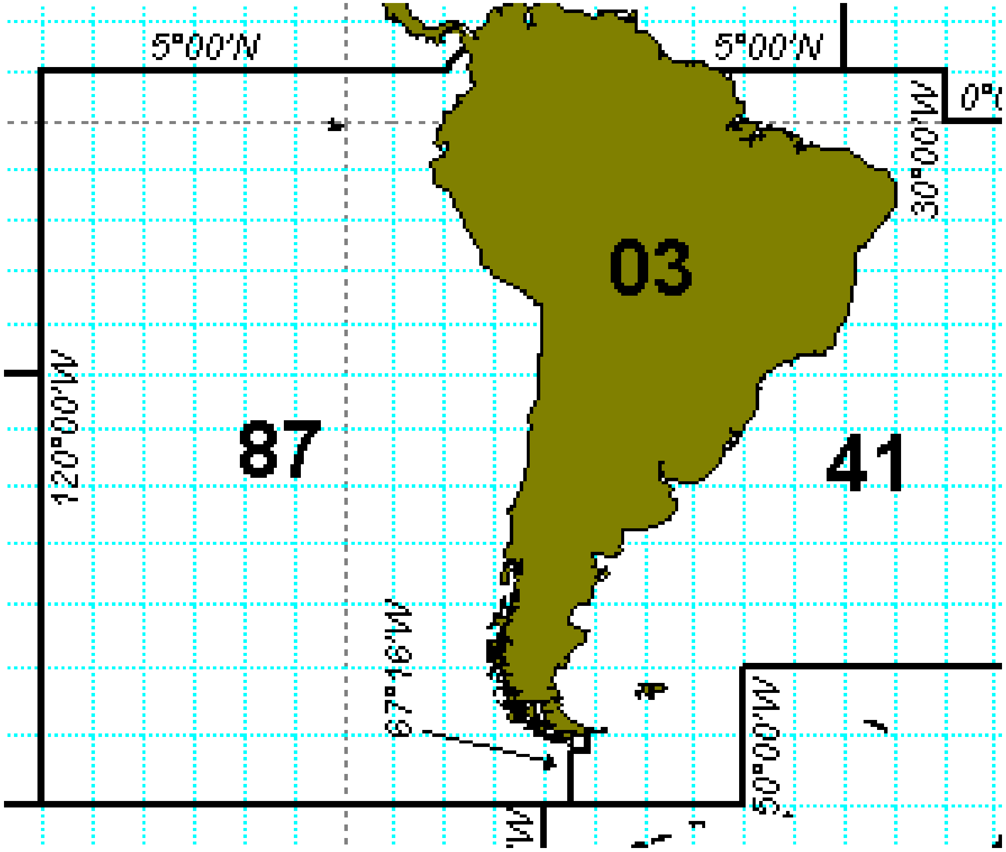
Stock assessment of jumbo flying squid (*Dosidicus gigas*) in the FAO Area 87 using SPiCT.

Ignacio Payá, july 2026

A single stock hypothesis

Stock area = FAO 87 area:

1. **SPFRMO area**
2. **Ecuador EEZ**
3. **Perú ANJ**
4. **Chile EEZ**



Background

- The recommendation for short-lived squid is to do an in-season stock assessment and in-season management.

- So far in-season stock assessment seems to be hard to implement in squid (*Dosidicus gigas*) in the SPRFMO area.

- An alternative method is to apply a surplus production model.

SPiCT

- SPiCT is considered as the assessment model for several stocks in ICES (KMSYSPICT 2021).
- SPiCT: Stochastic Production model in Continuous Time (Pedersen & Berg, 2017).
- Based on generalised surplus production models in the form of Pella & Tomlinson (1969)
- State-space models that separate random variability of stock dynamics from error in observed indices of biomass.
- Enables error in the catch process to be reflected in the uncertainty of estimated model parameters and management quantities.
- Ability to provide estimates of exploitable biomass and fishing mortality at any point in time from data sampled at arbitrary and possibly irregular intervals.

SPiCT

- Biomass and fishing mortality are unobserved processes = random effects = process errors

- Catches and abundance indices are observed variables = observation errors

Background

SPiCT applied to D. gigas in FAO Area 87

	Data up to	Years to SC	Years to Commission	SPRFMO	WD
Payá 2022	2020	**2**	3	SC10	SC10-SQ10
Payá 2023	2021	**2**	3	SC11	SC11-SQ07
Payá 2024	2022	**2**	3	SC12	SC12-SQ01
Payá 2025	2024	**1**	2	SC13	SC13 - SQ 06

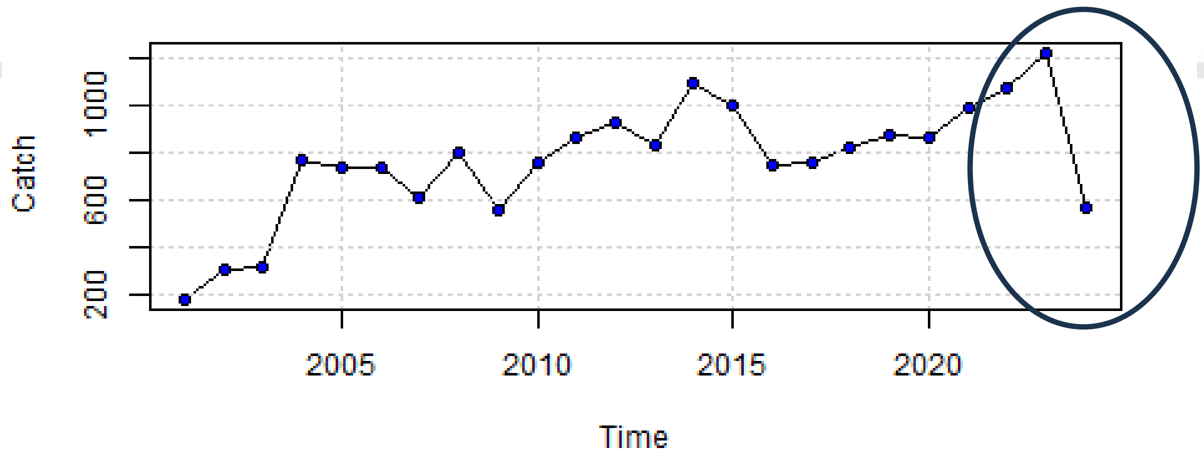
Last year improved the timing from model update to SC meetings.

Cases in 2025

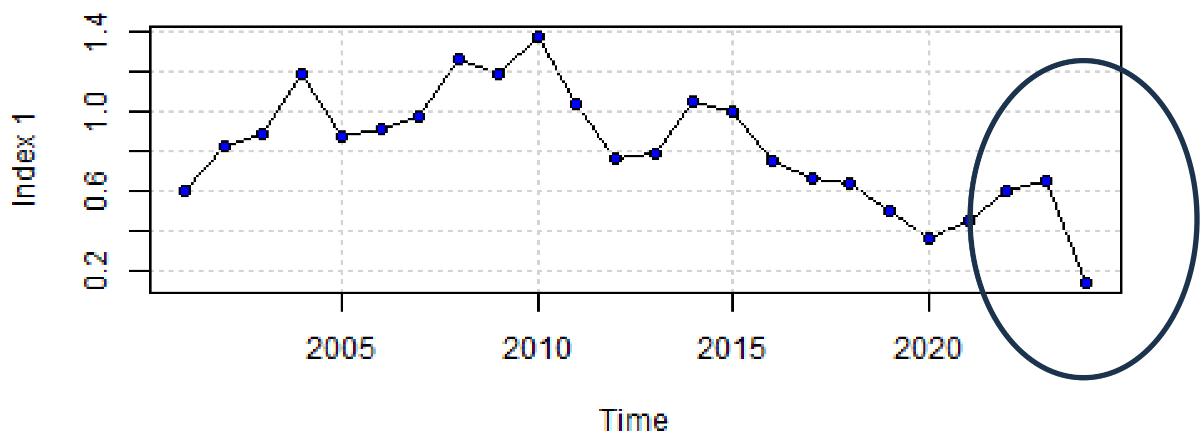
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
Abundance Indices	Global Index (average of country weighted by catch)	**6 Indices.** 1. China 1 2 China 2 3. Korea 4. China Taipei 5. Chile 6. Perú	Global Index (average of country weighted by catch)	**6 Indices.** 1. China 1 2 China 2 3. Korea 4. China Taipei 5. Chile 6. Perú	Global Index (average of country weighted by catch)	**6 Indices.** 1. China 1 2 China 2 3. Korea 4. China Taipei 5. Chile 6. Perú	**6 Indices.** 1. China 1 2 China 2 3. Korea 4. China Taipei 5. Chile 6. Perú
Shape of productive curve (“parameter n”)	Free	Free	Free	Free	**Schaefer Model**	**Schaefer Model**	Free
Parameter “r”	Free	Free	Prior (1, 0.6)	Prior (1, 0.5)	Prior (1, 0.5)	Prior (1.5, **0.15**)	Prior (1.5, **0.15**)

Initial year was 2001 in all cases

Catch and CPUE index up to 2024

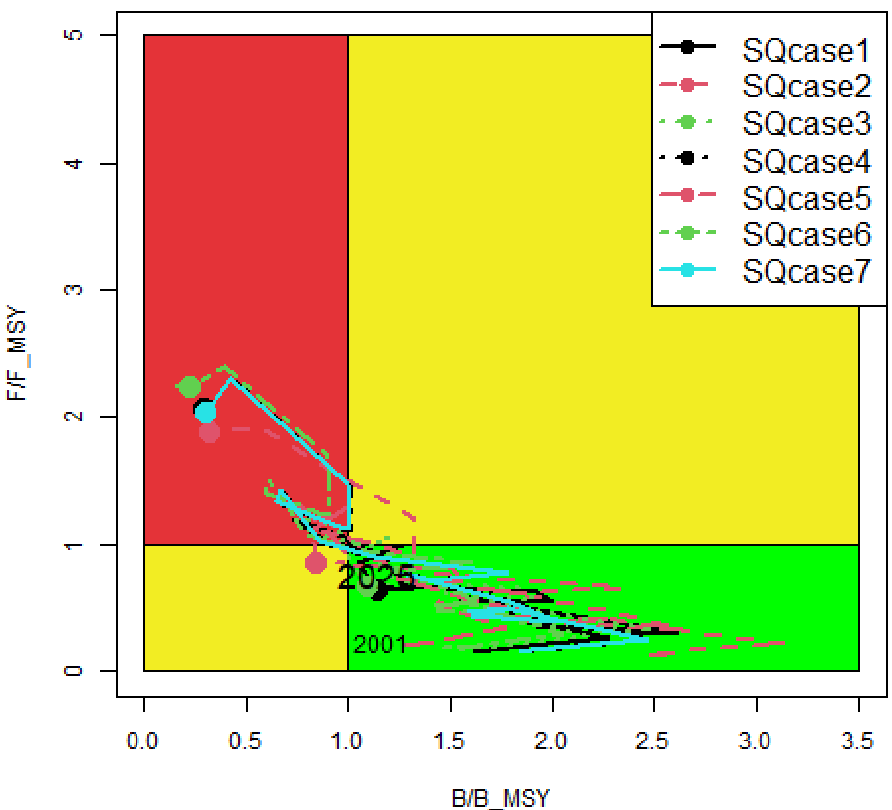


Nobs I: 24

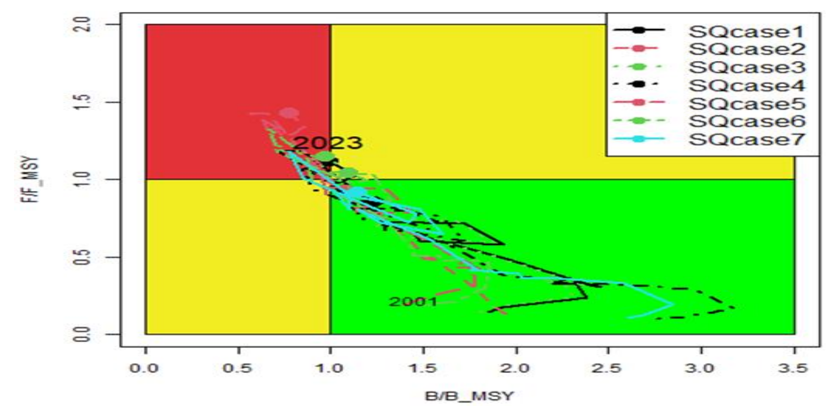


Stock status at 2024

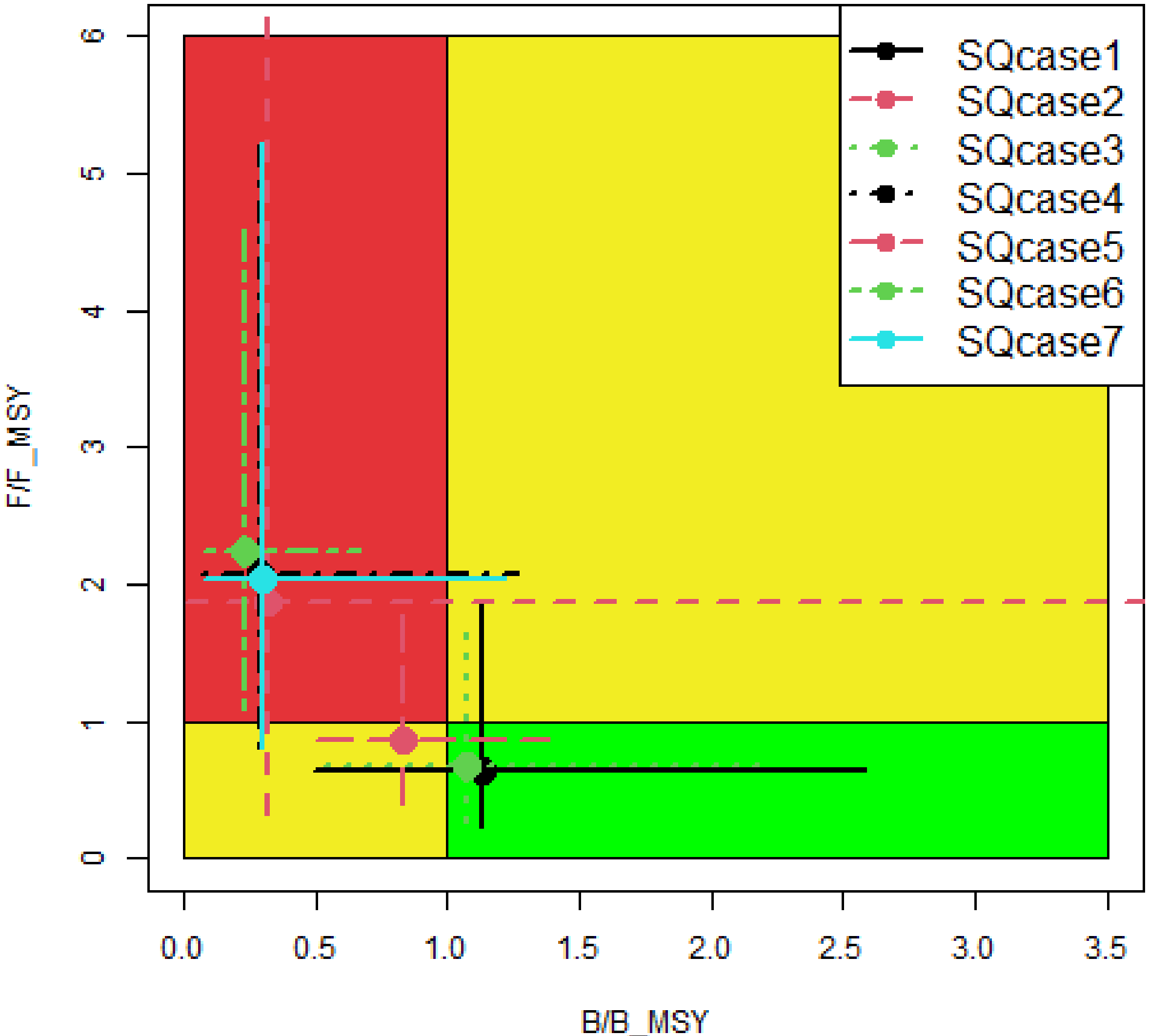
- Worse status than in the last assessment
- 2025 assessment



2024 assessment



Stock status at 2024



Conclusions in 2025

1. Strong decrease of catches in year 2024.
2. 2024 stock status was overexploited (B<Bmsy) and overfished (F>Fmsy).
3. Catch quotas (HCR) : 400-450 thousand tones.
4. MSY ~ 900 thousand tones
5. MSY ~ Recovery time is short (1.5-2 years).

Aims

To assess the D. gigas stock in FAO87 area using SPiCT *with data up to 2025* .

To evaluate the impact of:

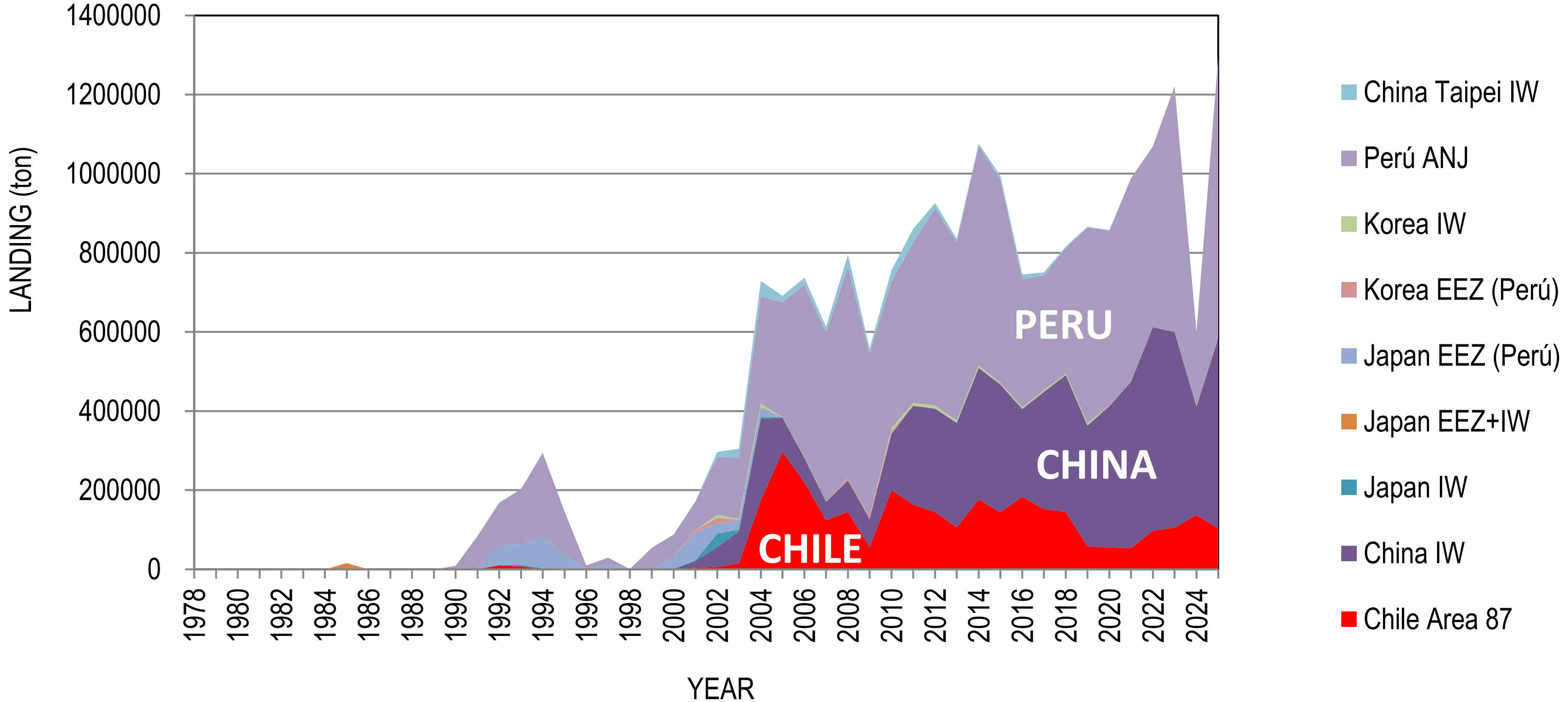
- fit the model to one or several indices.
- Different productivity curves.
- priors for “r”

DATA

CATCHES BY COUNTRY

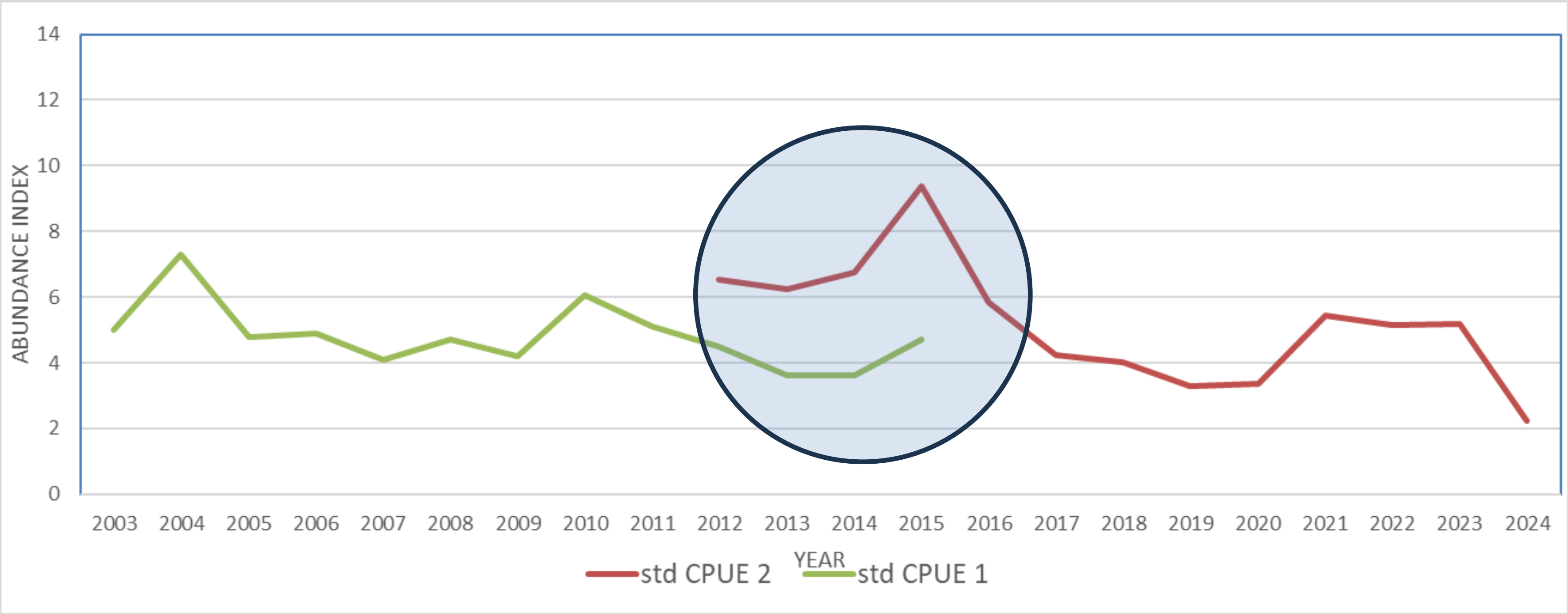
Whole catch recovered in 2025 because of Perú and China.

	2025					
	Chile	China	Korea	Peru	Chinese Taipei	ToTal
Catch (ton)	103667	480368	0	712155	231	1296421
Percentage	8%	37%	0%	55%	0.0%	100%



Reduce the timing between stock assessment and management meeting

China abundance indices used in 2025 assessment

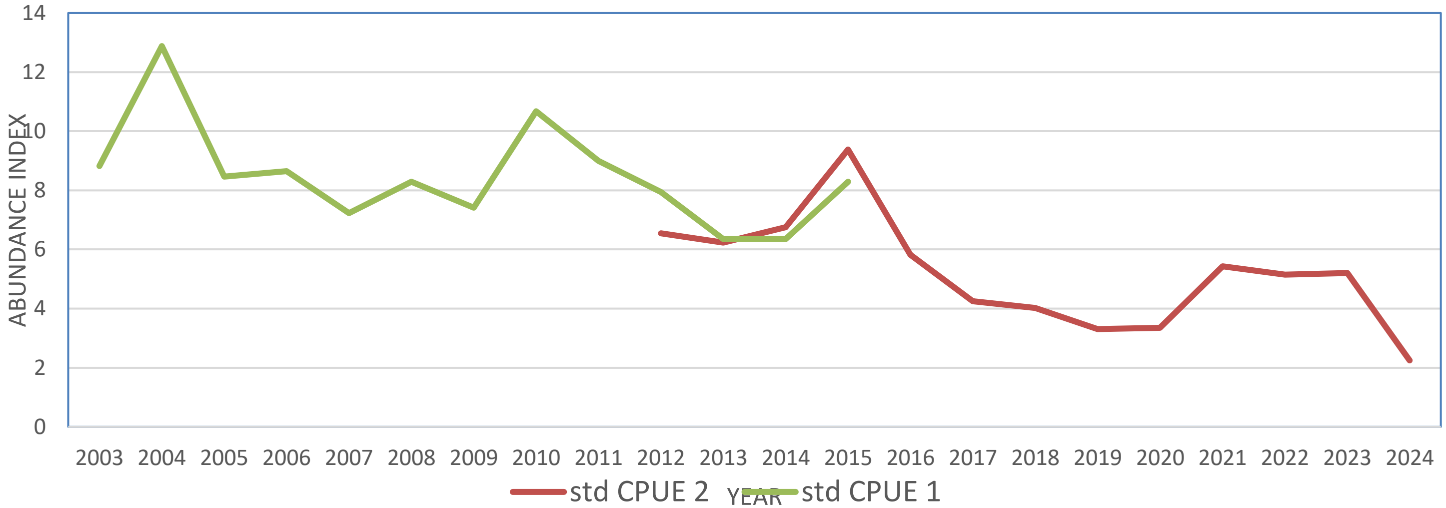


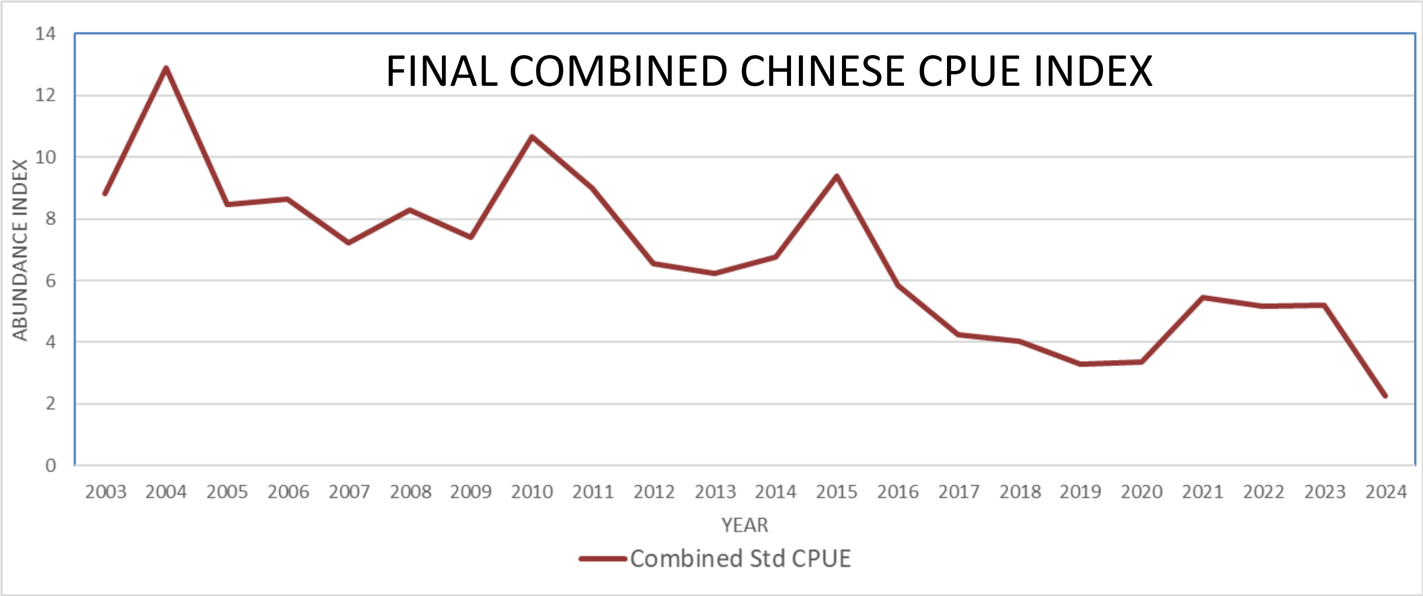
Standardized CPUE China 1

5th Meeting of the Scientific Committee
Shanghai, China 23 - 28 September 2017
SC5-SQ02
A Stock assessment of the jumbo flying squid (*Dosidicus gigas*) in Southeast Pacific Ocean (2017)

Standarized CPUE China 2 (UPDATED AT 2024 by DR. Gang Li, 2025)
12th MEETING OF THE SCIENTIFIC COMMITTEE
30 September to 05 October 2024, Lima, Peru
Updated stock assessment of the jumbo flying squid in the South-East Pacific
Gang Li, Yangming Cao, Xinjun Chen
Shanghai Ocean University

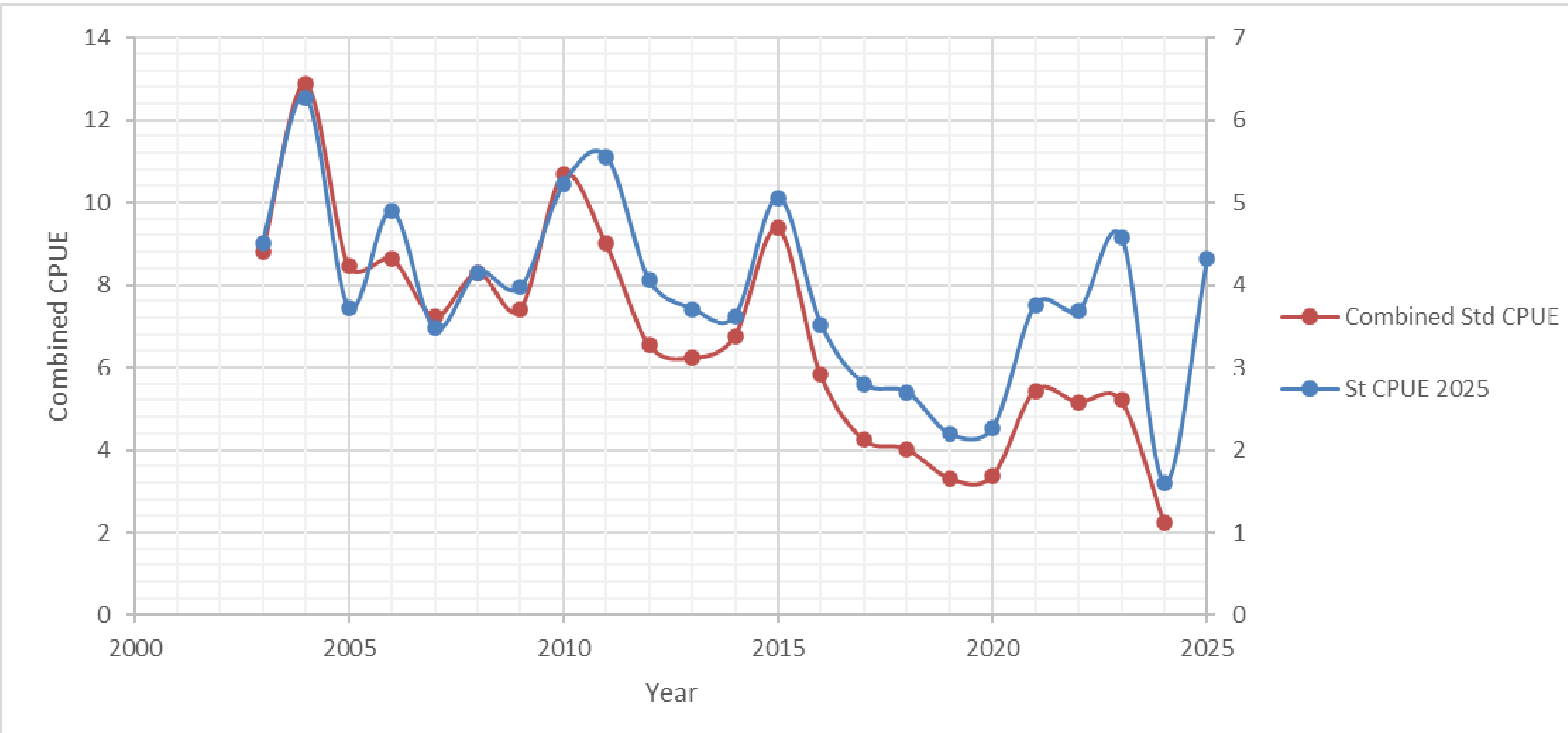
China abundance indices used in 2025 assessment





China CPUE up to 2025

Combined Chinese CPUE replaced by an annual average of standardized CPUE (provided by Dr. Gang Li) by month



Abundance index in Perú

Artisanal boat CPUE.

CPUE weighted by fishing areas

Without GLM

IMARPE 2025. Indicadores biológicos, pesqueros y poblacionales del calamar gigante Dosidicus gigas y perspectivas de pesca para el 2025. Informe Técnico. https://cdn.www.gob.pe/uploads/document/file/8143487/6819809-indicadores-biologicos-pesqueros-y-poblacionales-del-calamar-gigante.pdf

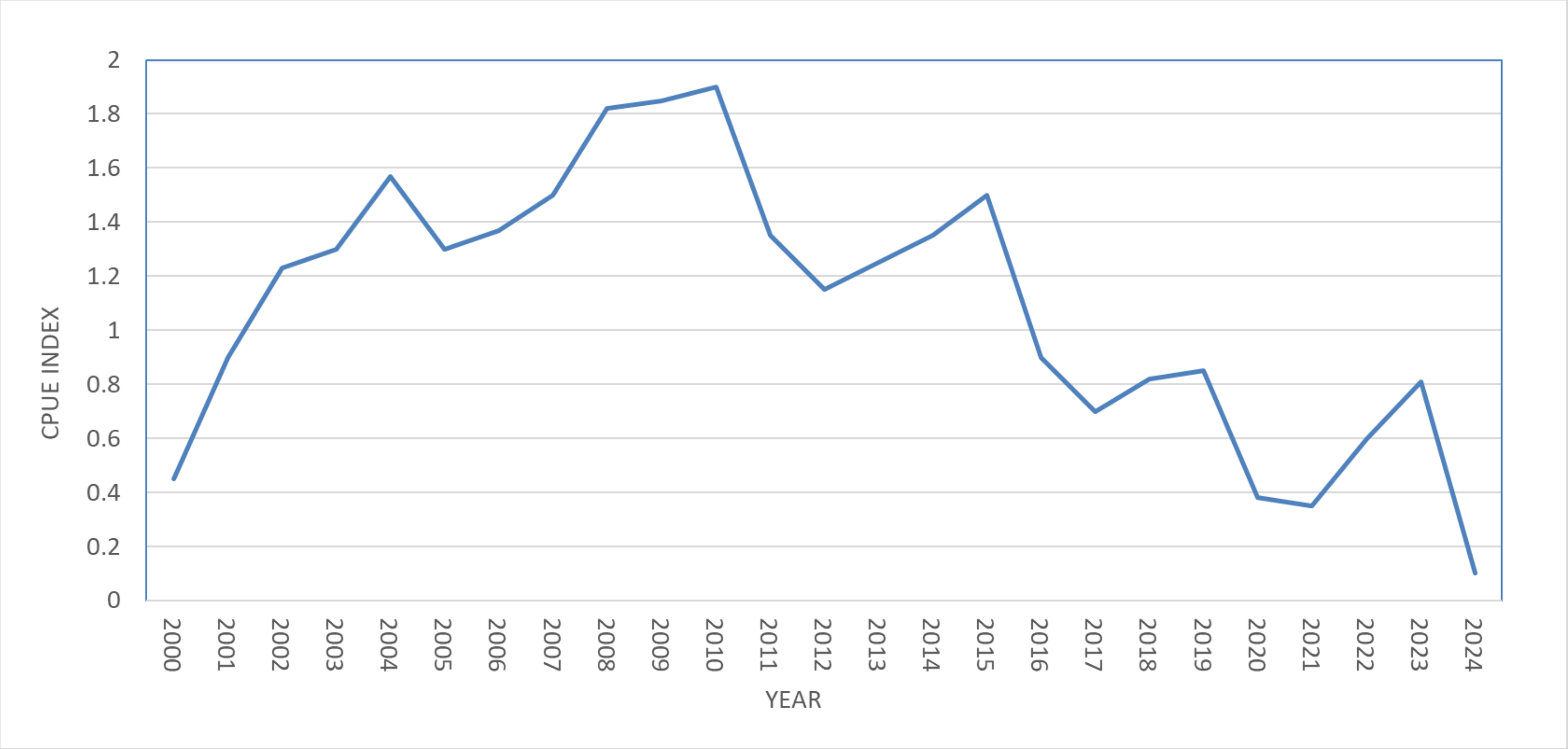
Gulland (1971) method

$$CPUE_{eff} = \frac{C}{\tilde{f}}$$

$$\tilde{f} = \frac{C}{qD} = \frac{A \sum C_i}{\sum A_i * C_i / f_i}$$

CPUE index in Perú up to 2024

No CPUE index for year 2025



Abundance index in Chile

standardized CPUE of artisanal boats (data up to 2025)

GLMM: Generalized Linear Mixed Model

CPUE= ton/trip

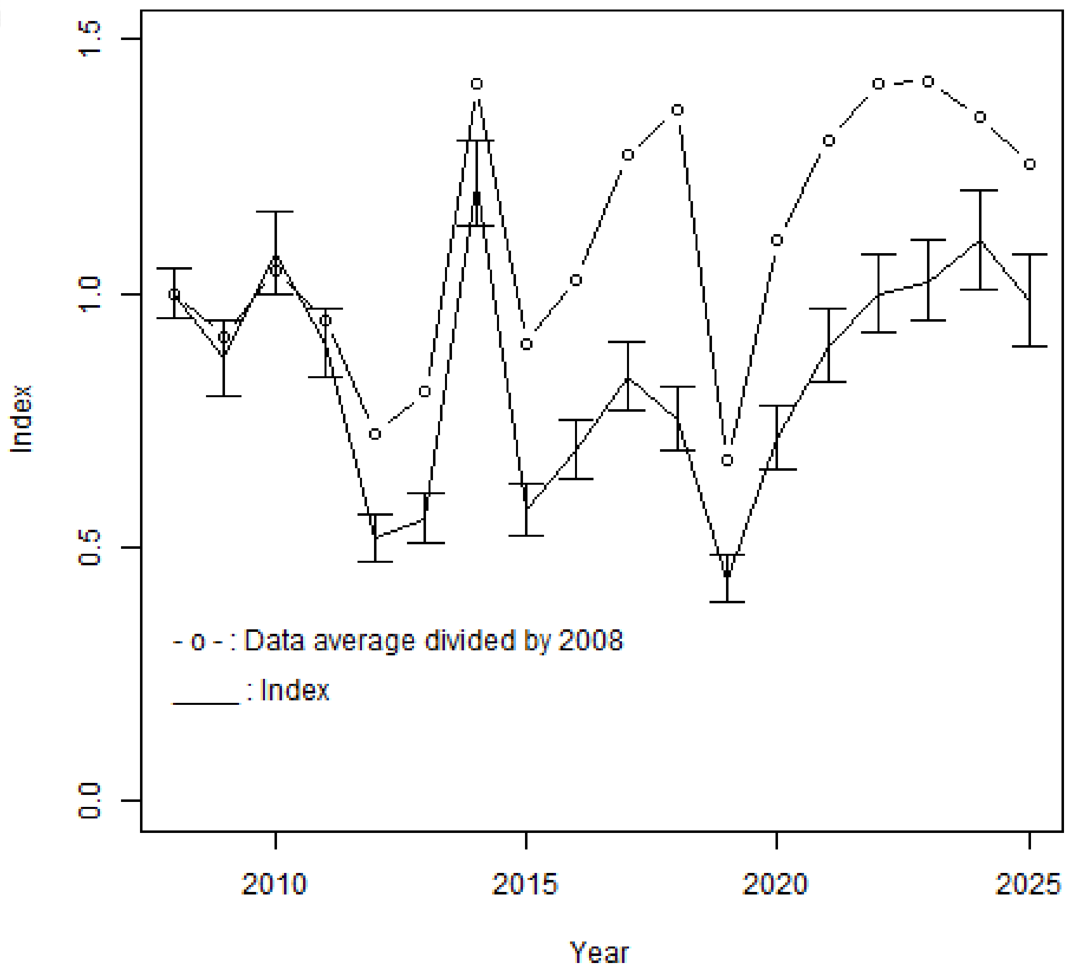
$CPUE^{1/3}$ ~ Year + Month + Region + (1 | Boat)

Fixed Effects = Year, Month and Region.

Random effects = Boat

Payá I. 2026. Estatus y posibilidades de explotación biológicamente sustentables de los principales recursos pesqueros nacionales, año 2027: Jibia. (Stock status and TAC 2026 for Chilean squid *Dosidicus gigas*

Abundance index in Chile

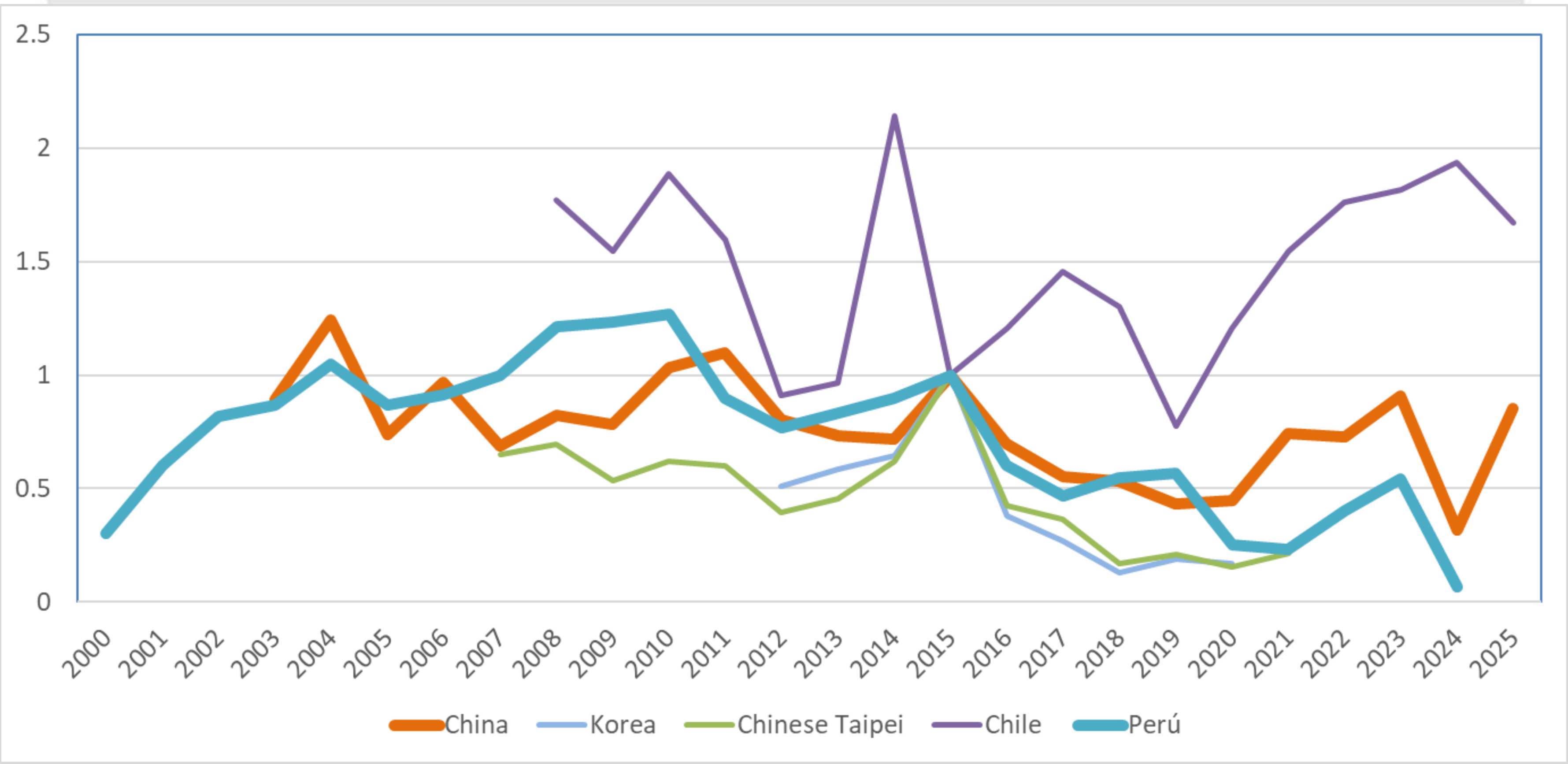


China Taipei and Korea CPUE

Nominal CPUE (ton/ fishing day) available in national reports

Abundance indices by country

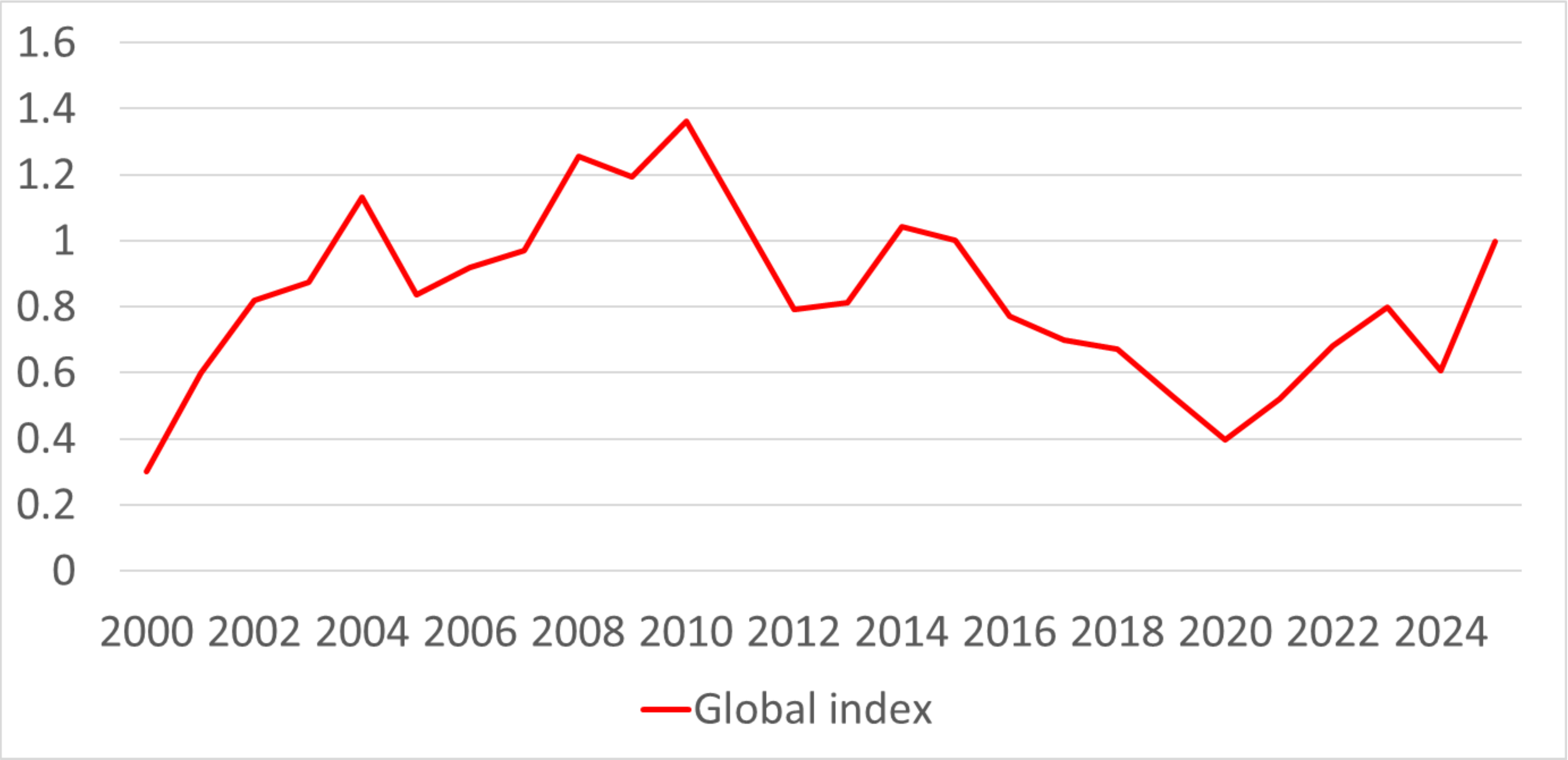
Indices / 2015 index



Global abundance index

Step 1. Indices / 2015 index

Step 2. Global = Average of relative indices weighted by catch



r Prior

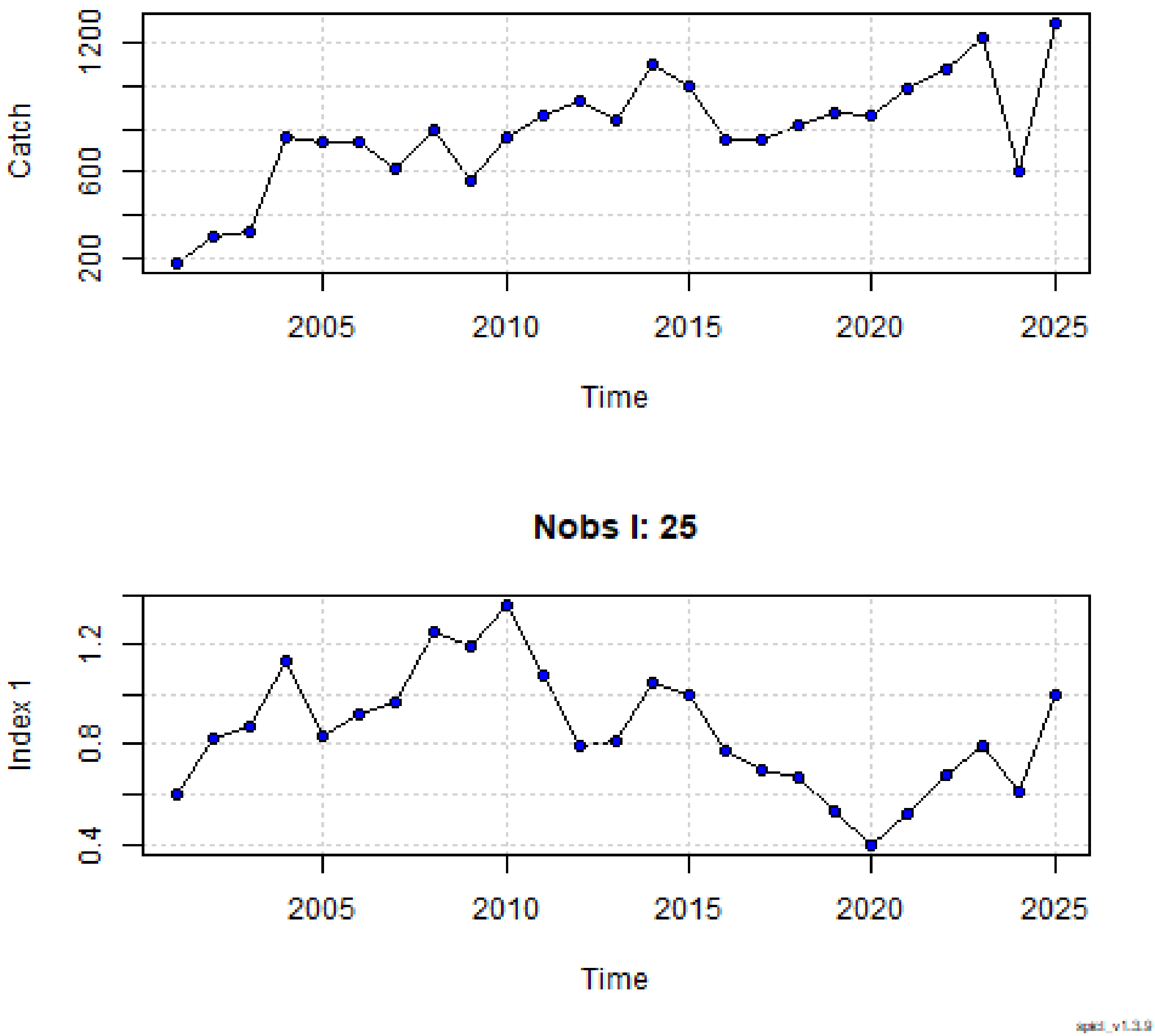
r	Species	Area	Reference
1.19	*Ommastrephes bartramii*		Ichii et al., 2002
0.85	*Illex illecebrosus*		Ichii et al., 2002
0.53 (0.16-1.71)	*Illex coindetii and Todaropsis eblanae*	Northwest Iberian Peninsula	Larivain et al., 2021
1.71-1.90	*Ommastrephes bartramii*	Northwest Pacific	Wang et al. 2016
1.23 – 1.68	*D. gigas*	México	Urías-Sotomayor et al 2018
1.33	*D. gigas*	Perú	Csirke et al., 2015

Cases

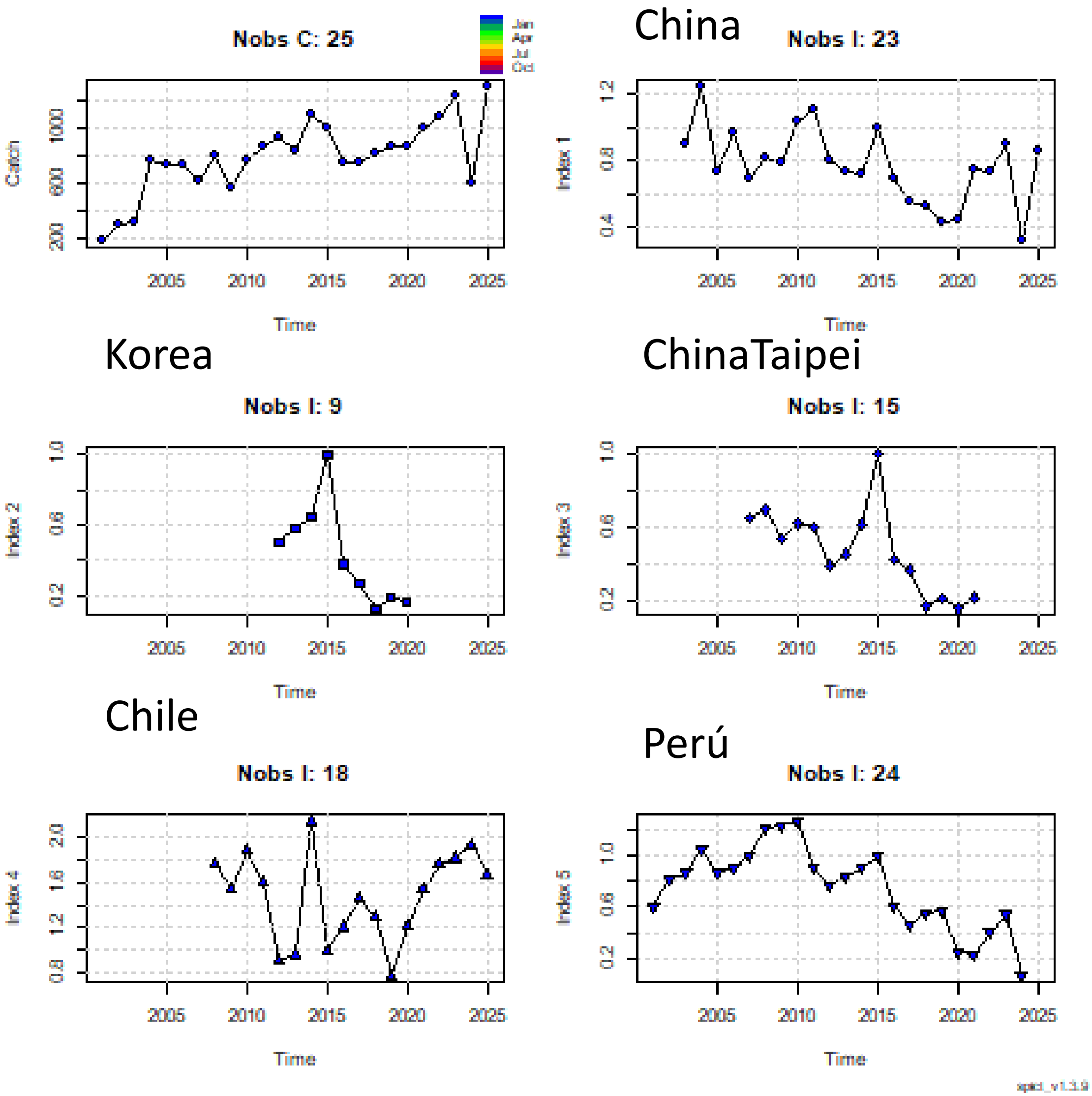
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
Abundance Indices	Global Index (average of country weighted by catch)	5 Indices.	Global Index (average of country weighted by catch)	5 Indices.	Global Index (average of country weighted by catch)	5 Indices.	5 Indices.
		1. China		1. China		1. China	1. China
		2. Korea		2. Korea		2. Korea	2. Korea
		3. China Taipei		3. China Taipei		3. China Taipei	3. China Taipei
		4. Chile		4. Chile		4. Chile	4. Chile
		5. Perú		5. Perú		5. Perú	5. Perú
Shape of productive curve (“parameter n”)	Free	Free	Free	Free	Schaefer Model	Schaefer Model	Schaefer Model
Parameter “r”	Free	Free	Prior	Prior	Prior	Prior	Prior
			(1, 0.5)	(1, 0.5)	(1, 0.15)	(1.5, 0.15)	(1.5, 0.3)

Initial year was 2001 in all cases

Input data: Case 1, 3 y 5



Input data: cases 2, 4, 6 y 7



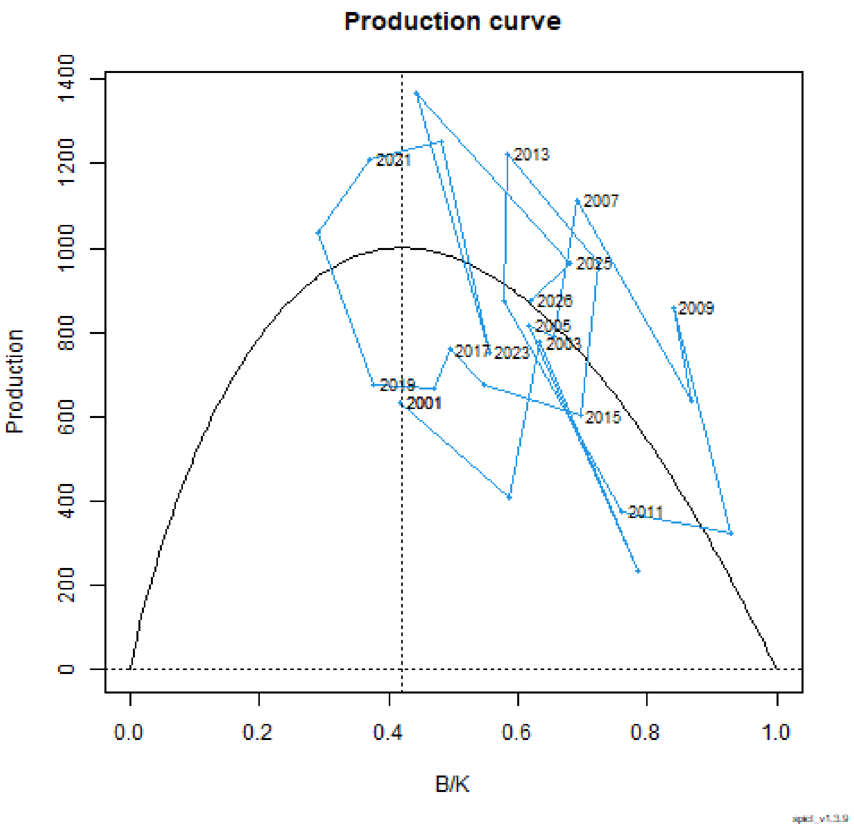
	Country	Type
1	China	Standardized
2	Korea	Nominal
3	China Taipei	Nominal
4	Chile	Standardized
5	Perú	Mean weighted by area

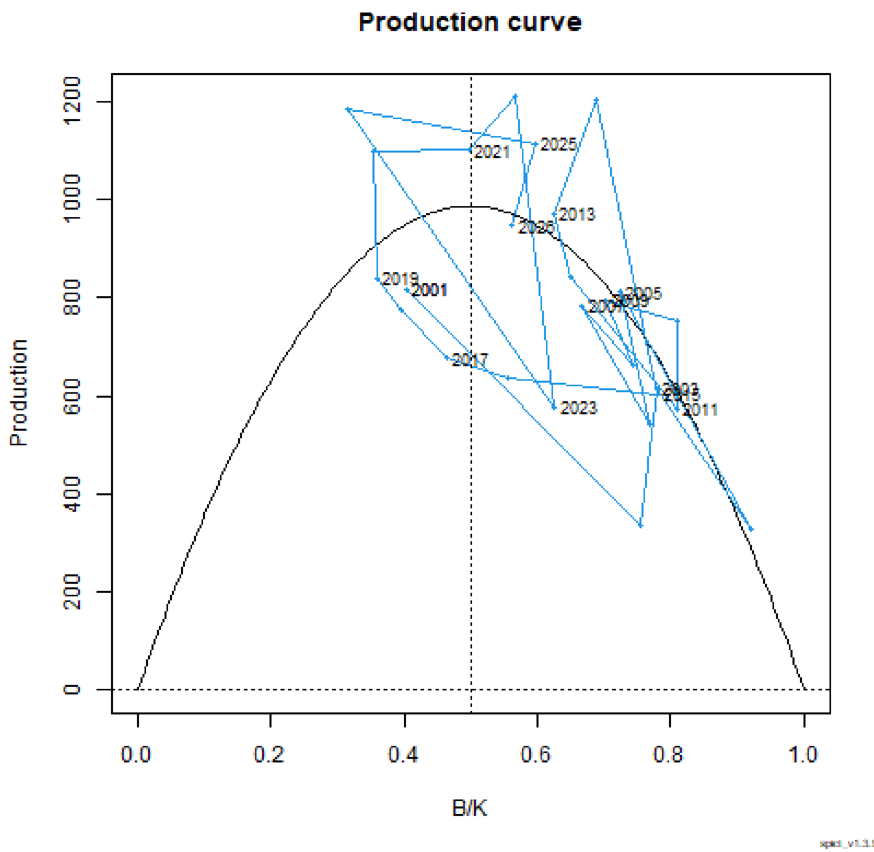
RESULTS

Production Curve

Case 1
Pella & Tomlinson
No Priors

Case 7
Schaefer
Prior r (1,5, 0.3)





Parameters by case

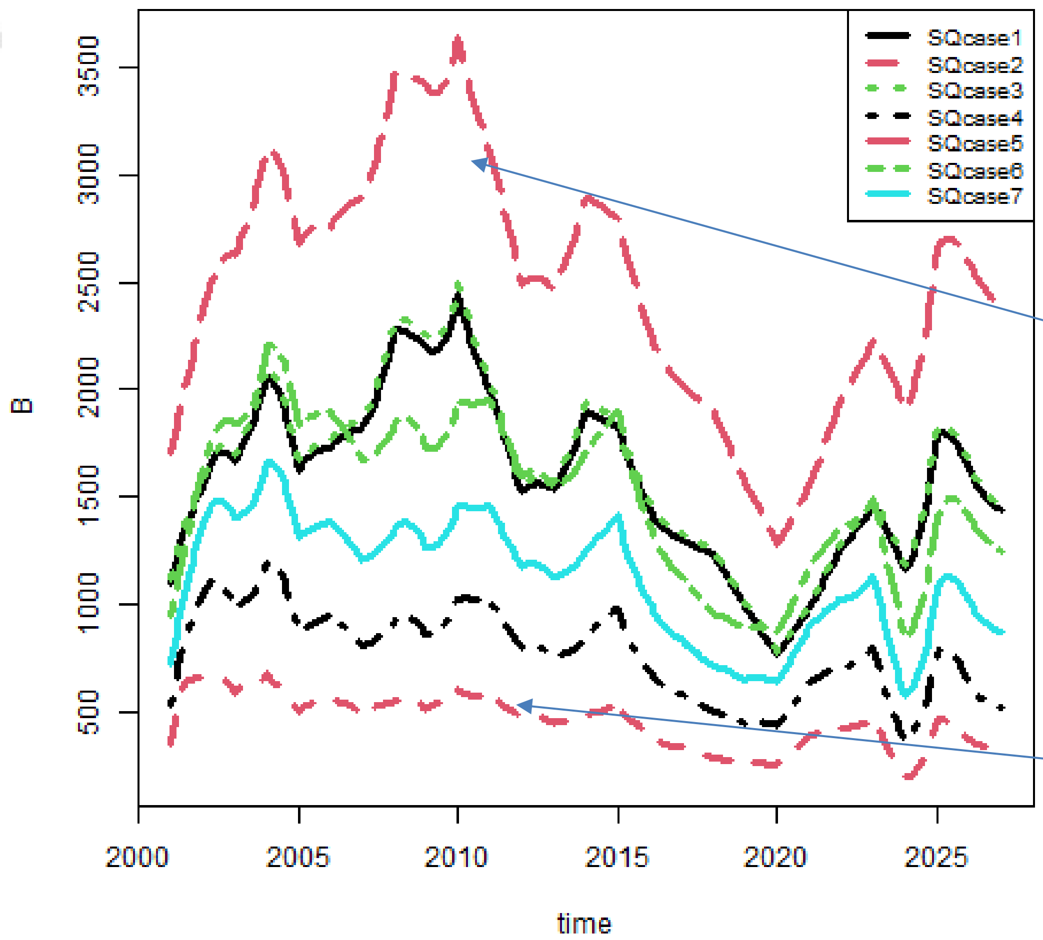
Parameters by case.

	1	2	3	4	5	6	7
r	1.2	6.32	1.07	1.82	1.04	1.65	2.18
rc	1.8	5.18	1.82	3.5	1.04	1.65	2.18
rold	3.63	4.39	6.16	41.5	1.04	1.65	2.18
K	2632	716	2755	1447	3879	2420	1810
Bmsyd	1109	386	1096	544	1940	1210	905
Fmsyd	0.9	2.59	0.91	1.75	0.52	0.82	1.09
MSYd	1001	999	998	951	1012	995	987

Parameters by case.

	1	2	3	4	5	6	7
r	1.31	2.53	1.19	1.75	1.11	1.68	1.6
rc	3.26	3.46	3.1	2.9	1.11	1.68	2.78
rold	6.64	5.49	5.2	8.47	1.11	1.68	10.8
K	1706	1236	1817	1560	3169	2213	1654
Bmsyd	560	543	583	629	1585	1106	651
Fmsyd	1.63	1.73	1.55	1.45	0.55	0.84	1.39
MSYd	914	941	904	913	877	929	905

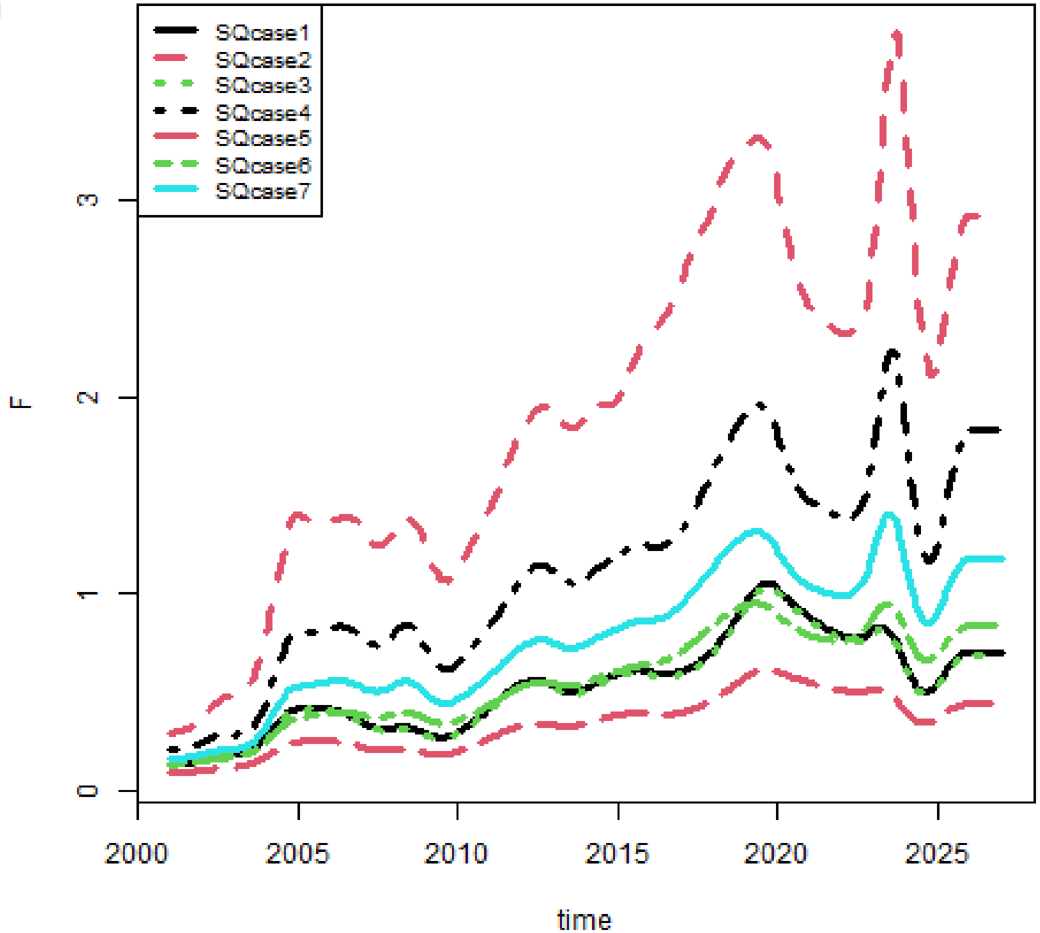
Biomass (ton/1000)



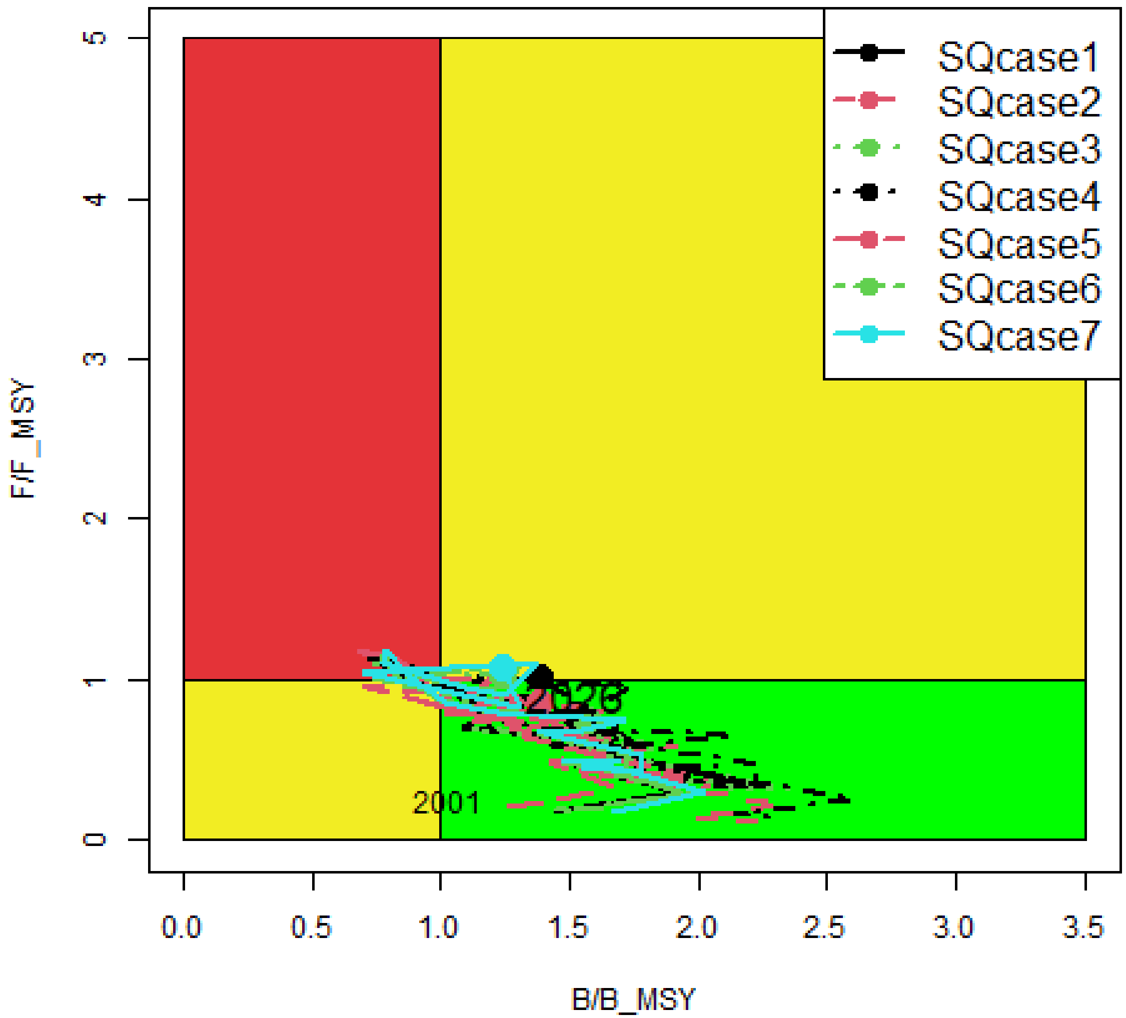
Symmetric productivity curve Schaefer model (Case 5)

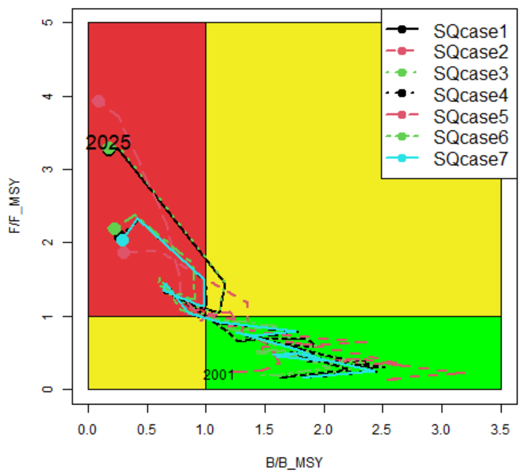
Non symmetric productivity curve Pella & Tomlinson (Case 2)

Fishing mortality rate by case

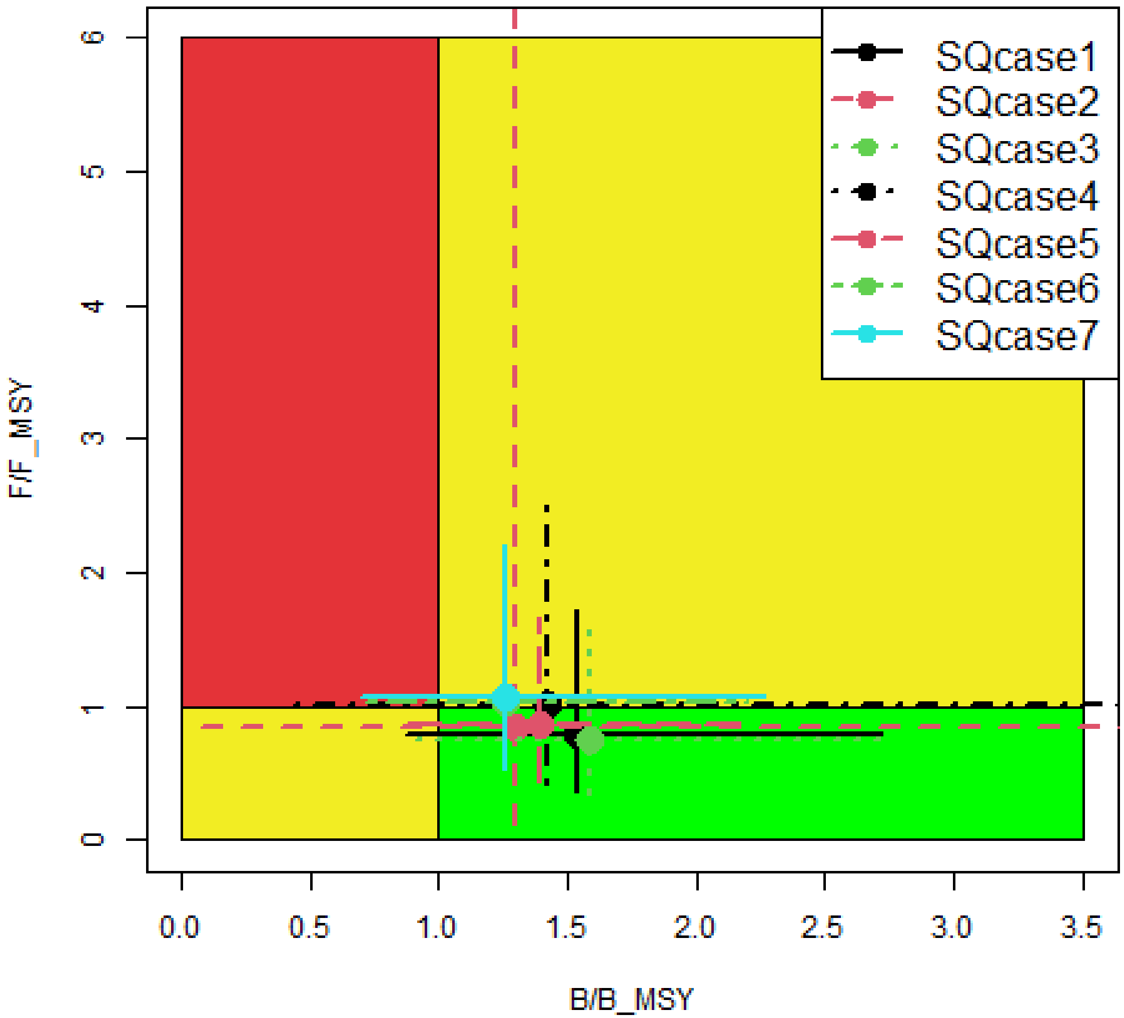


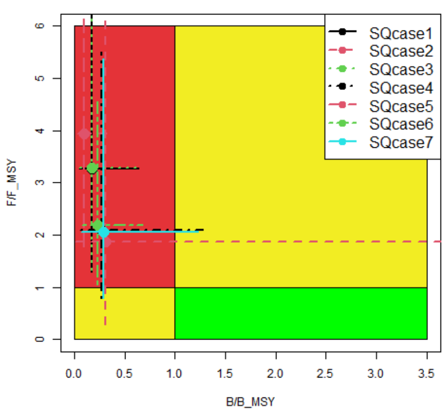
Stock status by case



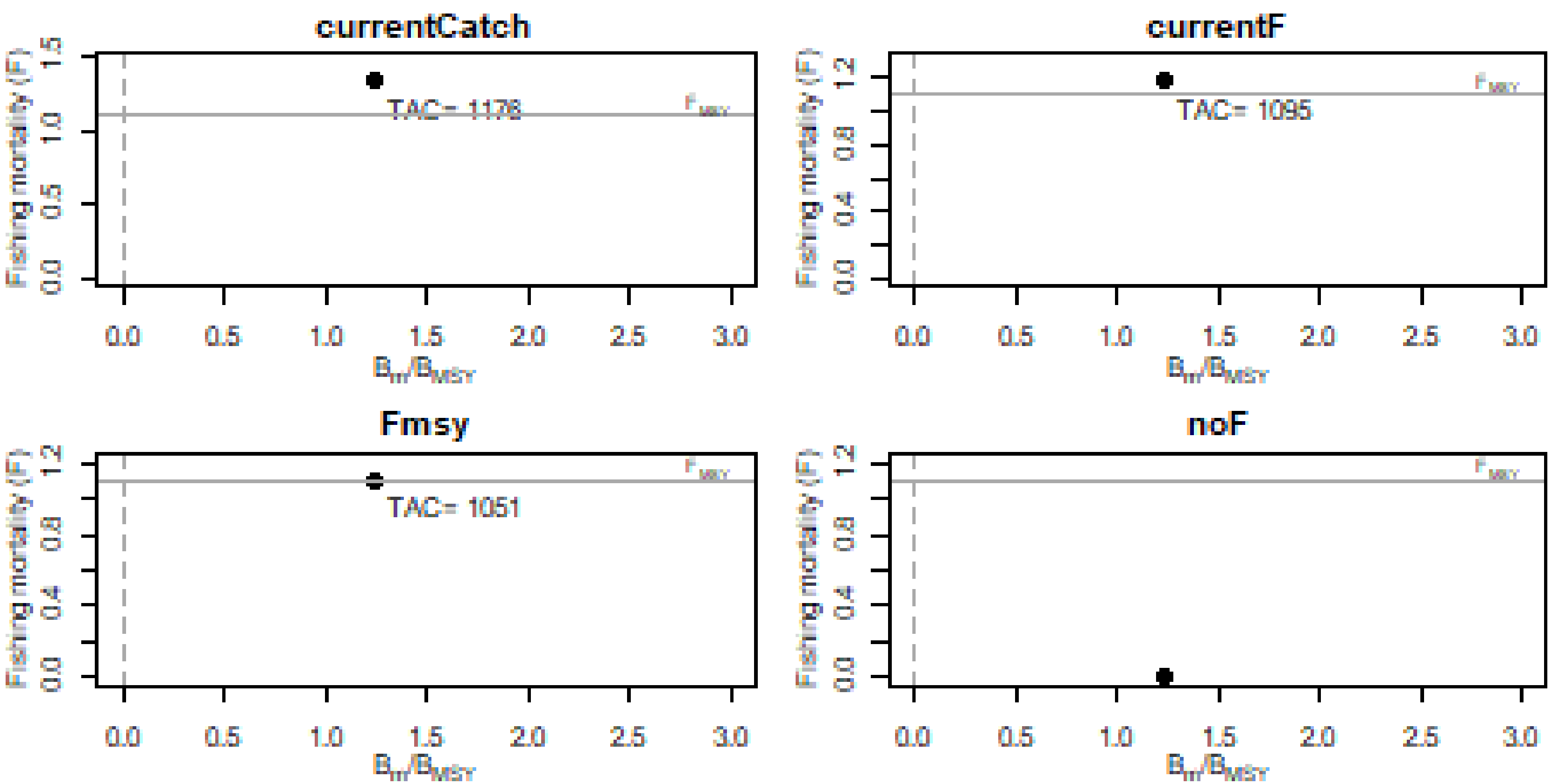


Stock status in 2025

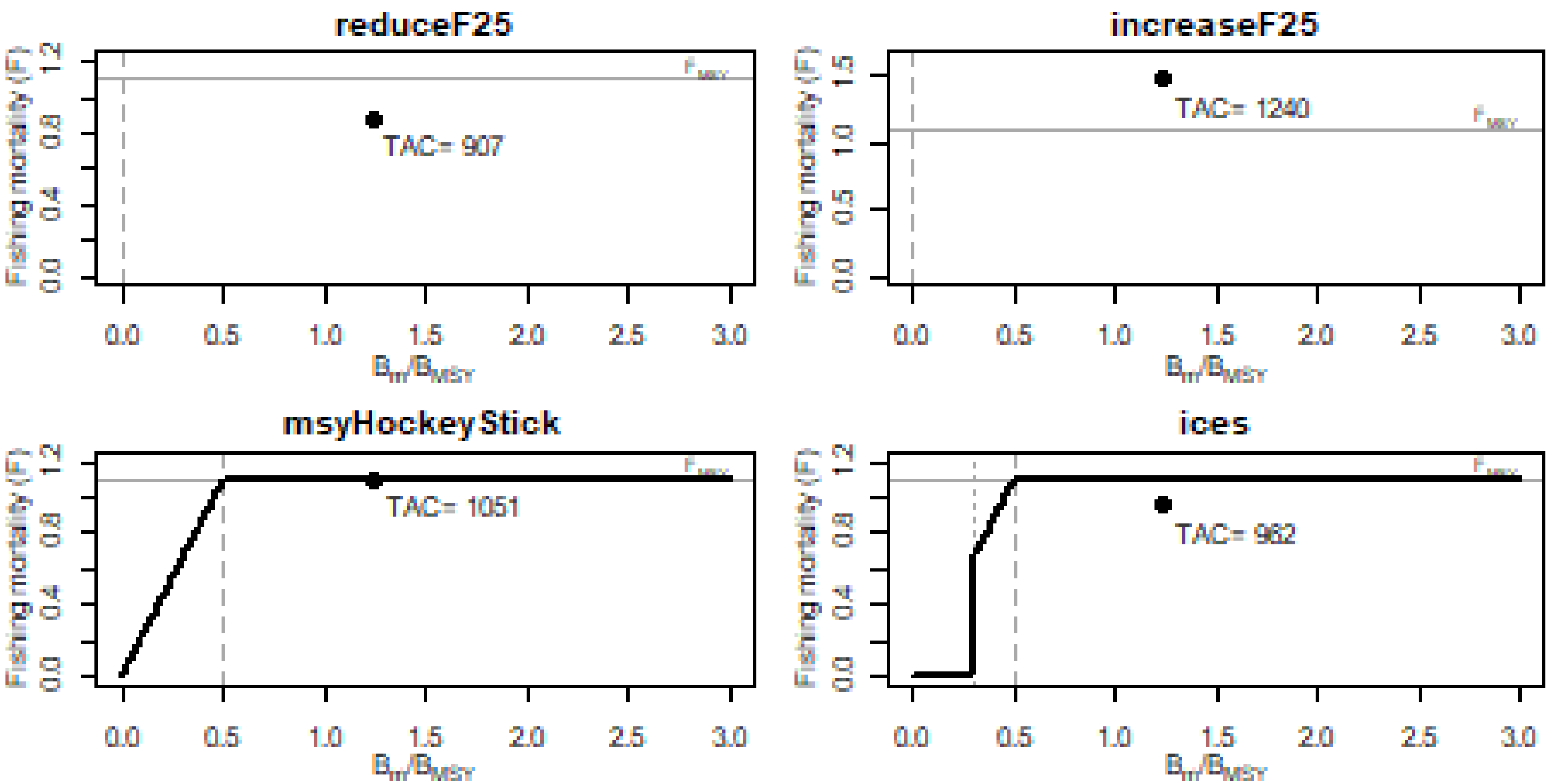




HCR (case 7)


currentCatch
TAC= 1178
F_MSY
Fishing mortality (F)
0.0
0.5
1.0
1.5
B_m/B_MSY
0.0
0.5
1.0
1.5
2.0
2.5
3.0
currentF
TAC= 1095
0.0
0.4
0.8
1.2
Fmsy
TAC= 1051
noF

HCR (case 7)



reduceF25
TAC= 907
increaseF25
TAC= 1240
msyHockeyStick
TAC= 1051
ices
TAC= 962
Fishing mortality (F)
0.0
0.4
0.8
1.2
0.5
1.0
1.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
B_{yr}/B_{MSY}
F_{MSY}

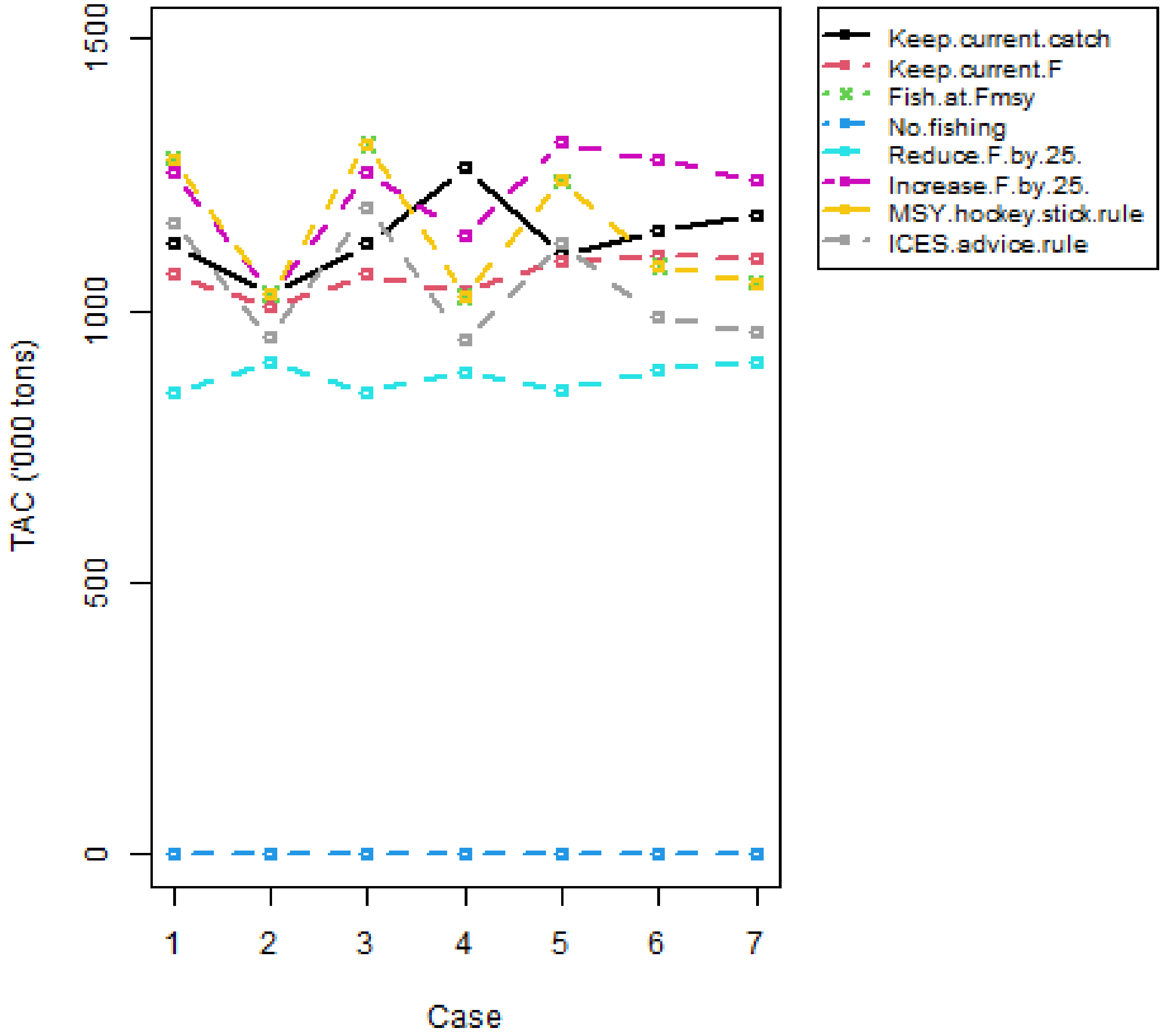
HCR by case

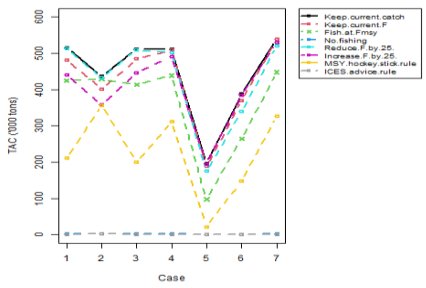
TAC (kt) by case.

	1	2	3	4	5	6	7
Keep.current.catch	1122	1032	1122	1264	1104	1146	1176
Keep.current.F	1066	1008	1066	1036	1092	1102	1095
Fish.at.Fmsy	1280	1032	1306	1026	1240	1082	1051
No.fishing	1	2	1	2	1	1	2
Reduce.F.by.25.	851	907	850	888	854	891	907
Increase.F.by.25.	1255	1032	1255	1137	1309	1278	1240
MSY.hockey.stick.rule	1280	1032	1306	1026	1240	1082	1051
ICES.advice.rule	1163	951	1187	947	1123	988	962

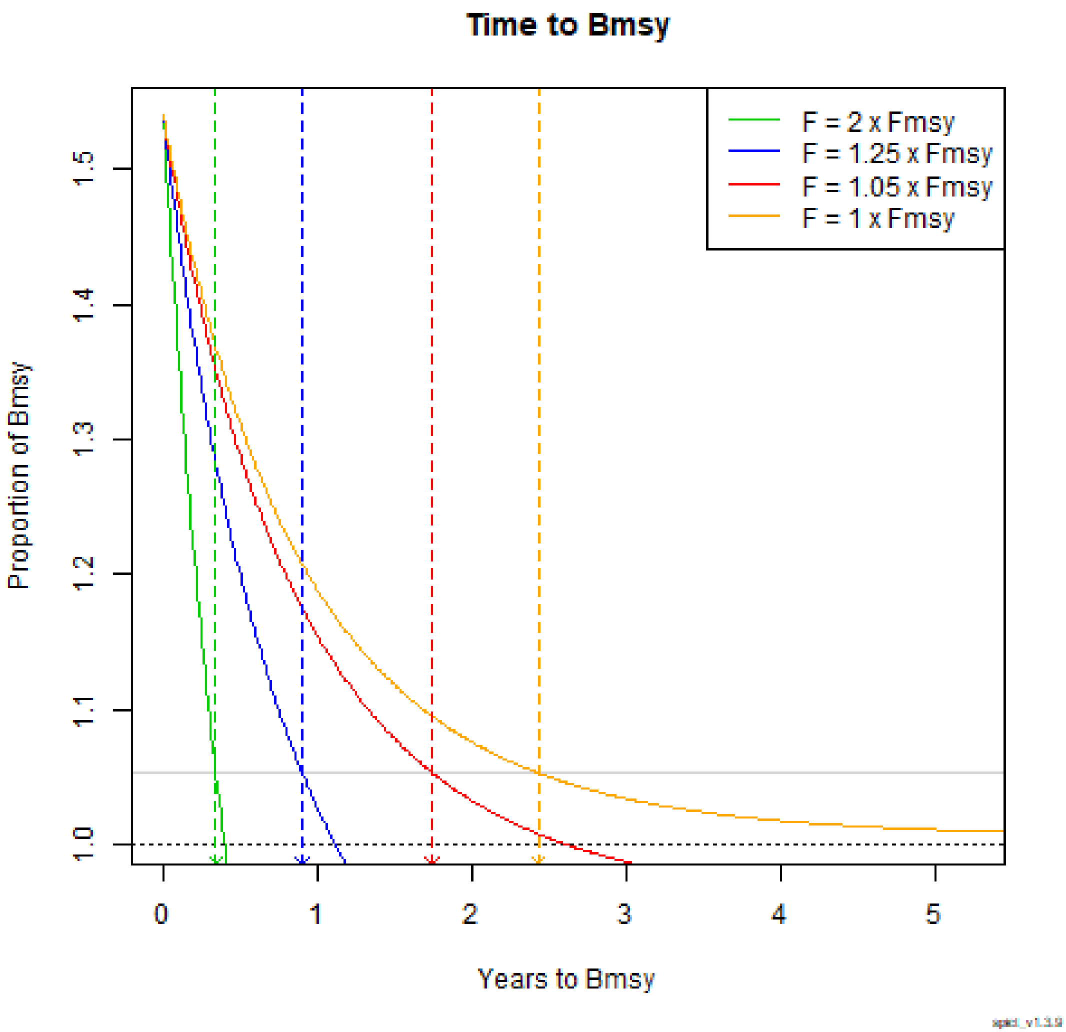
	1	2	3	4	5	6	7
Keep.current.catch	516	436	511	510	196	388	539
Keep.current.F	482	401	484	509	188	370	539
Fish.at.Fmsy	424	428	413	438	97	265	448
No.fishing	2	2	2	2	1	1	2
Reduce.F.by.25.	513	433	509	502	175	339	521
Increase.F.by.25.	440	358	447	490	194	384	529
MSY.hockey.stick.rule	210	357	201	311	22	148	326
ICES.advice.rule	0	2	0	0	0	0	0

TAC by HCR and case





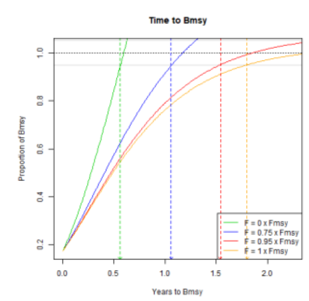
Time to Bmsy



CASE 1

Fmsy ->

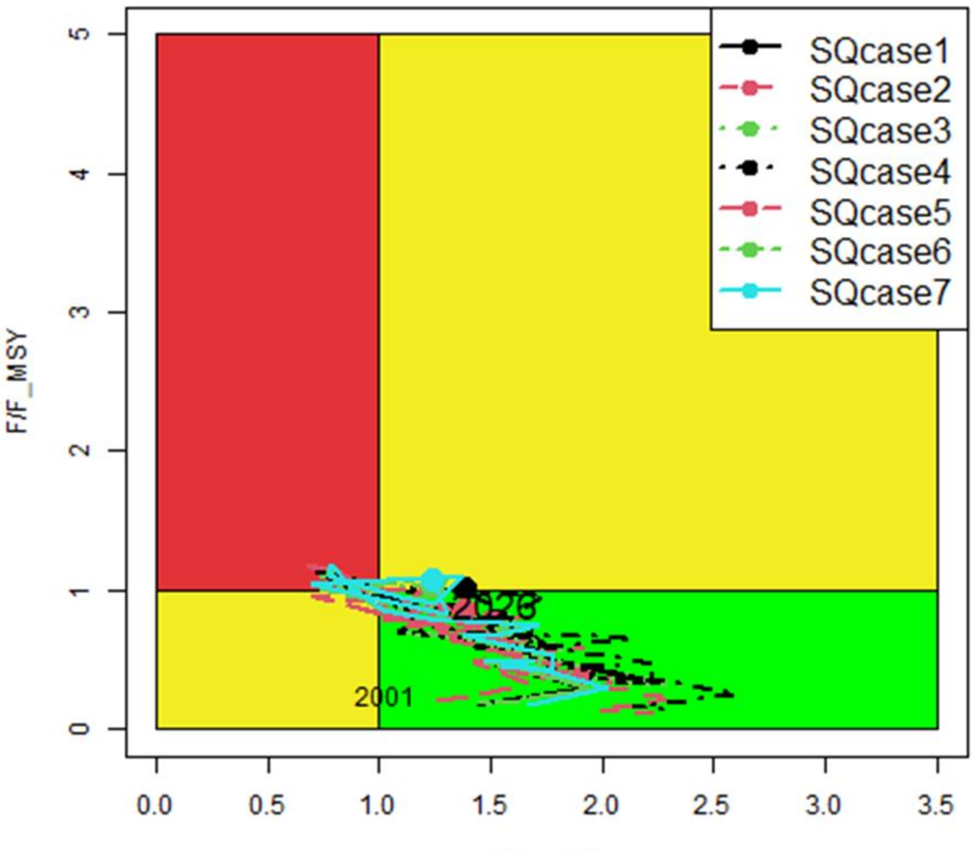
2.5 yrs to decrece to BMSY level



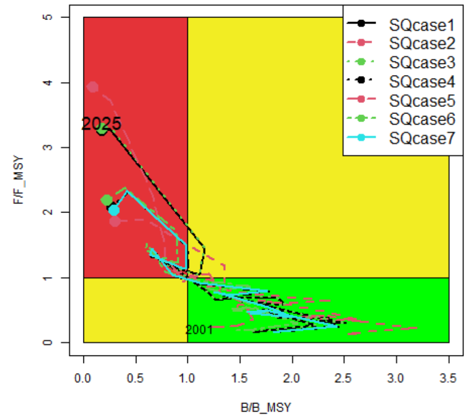
Discussion

Stock status is better than in the last stock assessment

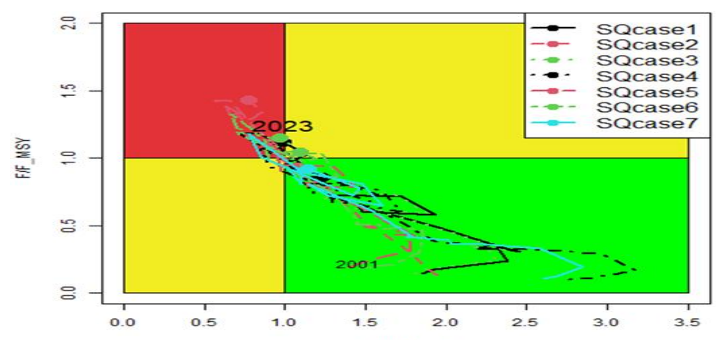
2026 assessment



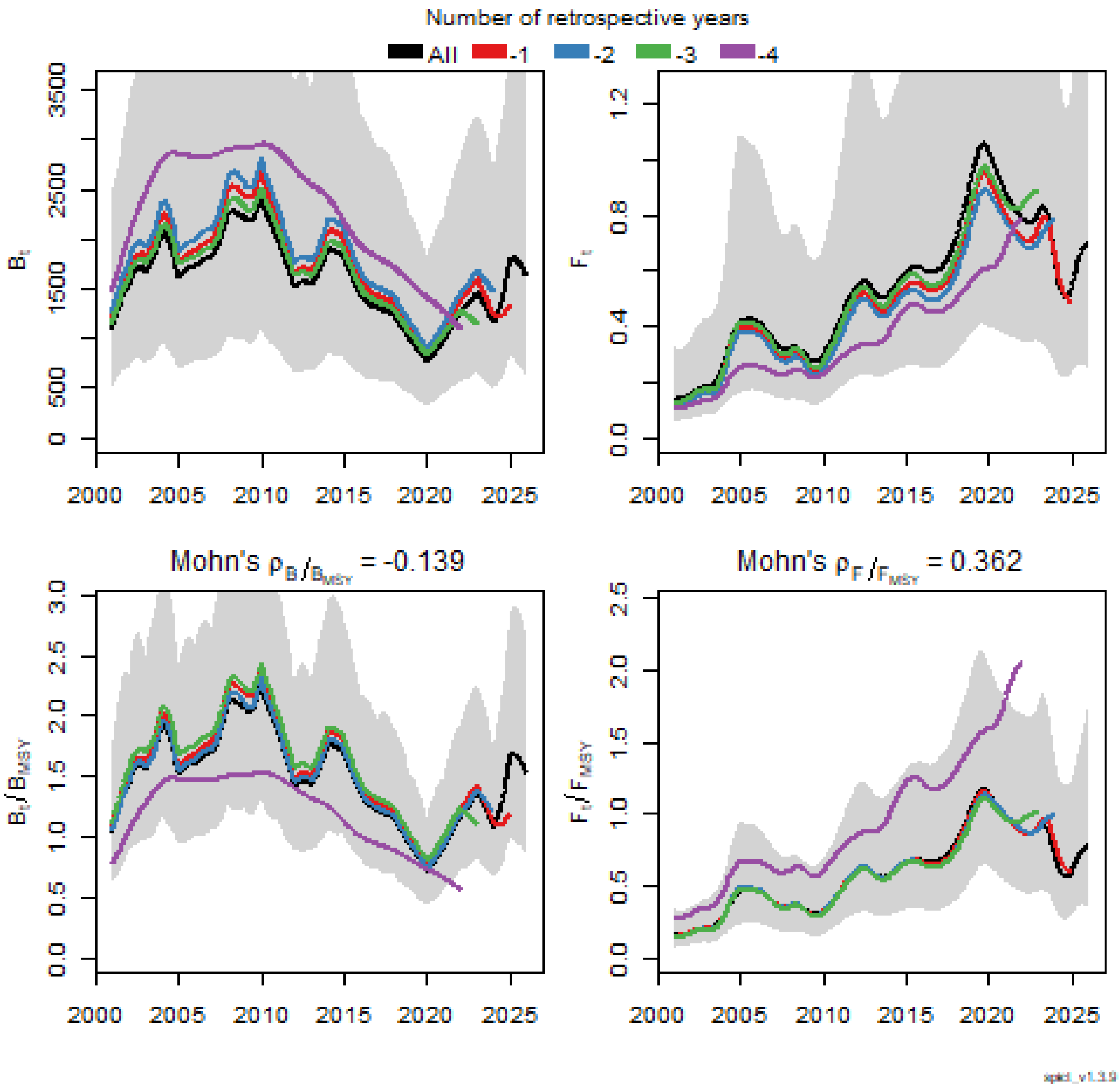
2025 assessment



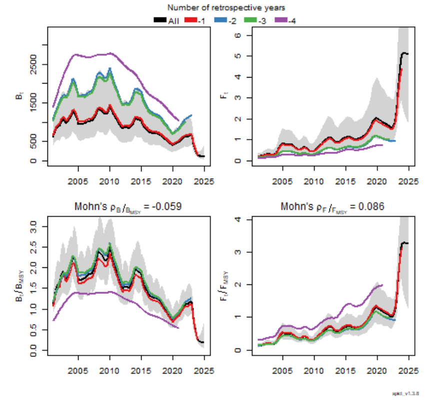
2024 assessment



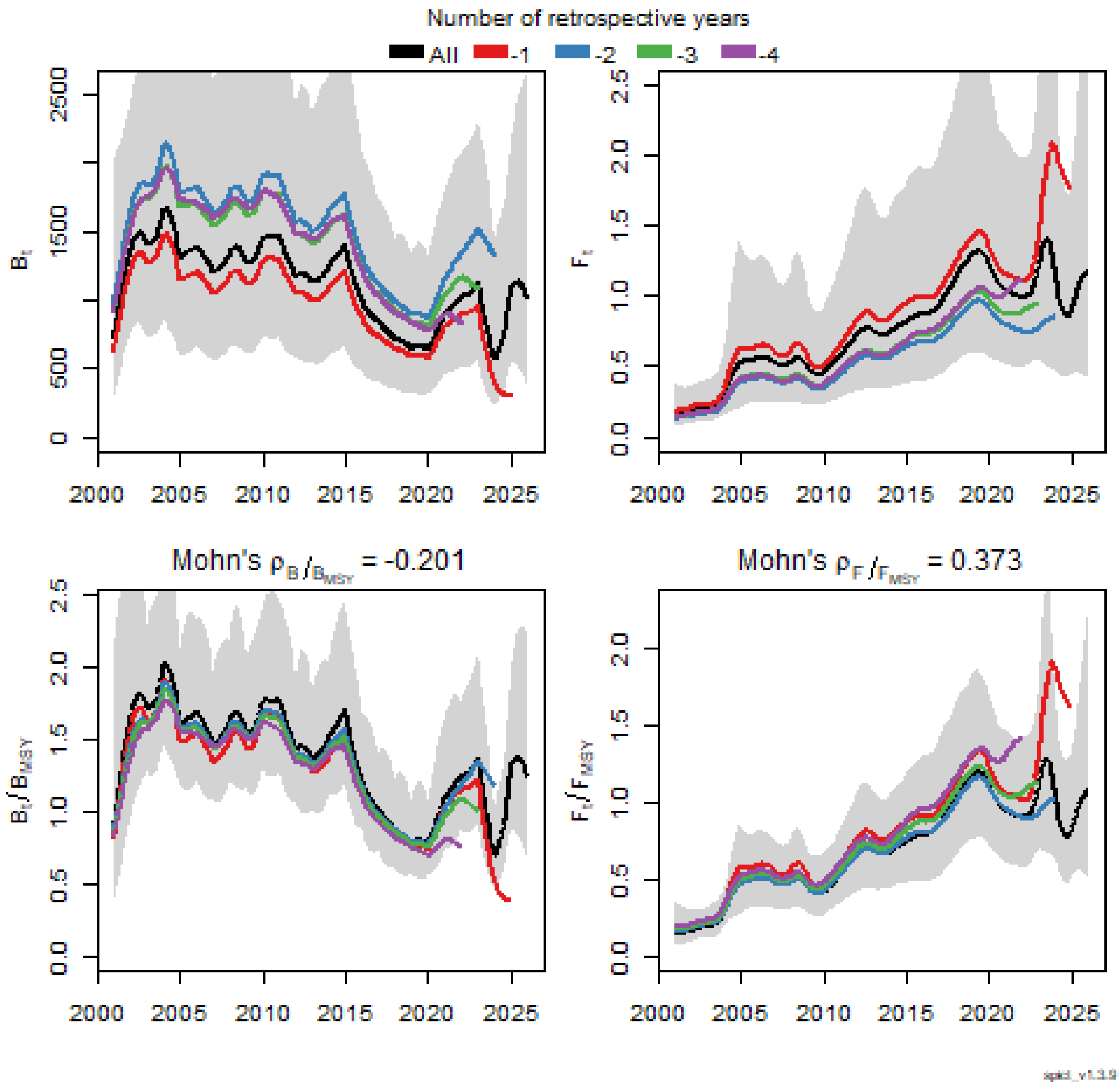
Retrospective Pattern



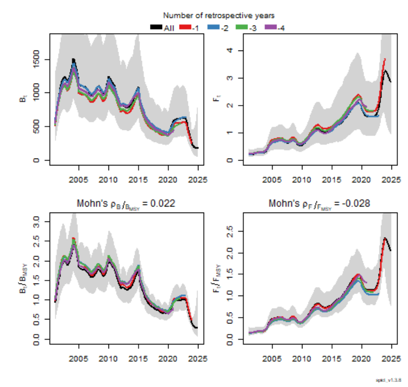
Case 1 (Global index)



Retrospective Pattern



Case 7 (global index)



Discussion

Catches were recovery in year 2025.

- 2025 whole catch was 2.1 times the 2024 whole catch
- 2025 Peruvian catch was 3.8 times the 2024 Peruvian catch.
- 2025 Chinese was 1.8 times the 2024 Chinese catch.
- 2025 Chilean was 0.8 times the 2024 Chilean catch.

CPUE indices were recovery in year 2025

- Chinese CPUE recovered in 2025
- 2025 Peruvian CPUE is not available yet.
- 2025 Chilean CPUE was similar to 2024 CPUE.

Discussion

SPiCT as a state-space model estimates **OBSERVATION ERRORS**

Abundance indices are key elements in SPiCT models.

- Historical global CPUE decreased until 2020 and then increased in the last 4 years.
- PERU abundance index is not available for 2025 (Gulland 1971 method)
- Preliminary Chinese CPUE index (Thanks to Dr. Gang Li).
- Chilean CPUE were similar during the last 3 years.
- Increase of Chinese CPUE during October-December is because of fishing fleet moves from equatorial fishing ground to off south of Perú fishing ground (large phenotype and good CPUE).

Discussion

SPiCT as a state-space model estimates **PROCESS ERRORS**

- Process error is **everything that is not included** in the model.
- **Biomass process errors** :
  - Interannual productivity variability
  - 3 phenotypes have different intrinsic growth rates and carrying capacities -> dissimilar productive curves.
  - Productivity variability related with environmental conditions.

Discussion

SPiCT as a state-space model estimates **PROCESS errors**

- Process error is **everything that is not included** in the model.
- **Fishing Mortality process errors**
  - Catchability by vessels
  - Catchabilities of different phenotypes
  - Catchabilities changes due to environmental variability.

Discussion

SPiCT as a state-space model estimates **PROCESS ERRORS**

- Question: How well the process errors can handle the phenotypes and environmental variability issue?
- Answer: Need to be check by simulation assuming different combination of phenotypes, environmental variables, etc.
- SQUIDSIM, squid simulation model (Payá 2019,2022,2023,2024 and 2025)
- SPRFMO 2024, new task team on squid simulation

Conclusions

1. Biomass estimated with Schaefer model were greater than with Pella & Tomlinson model
2. Model with one global CPUE were more stable and estimated greater biomass that models with 5 indices
3. Strong recovery of catches and CPUE in year 2025.
4. Quick recovery of stock status.
5. Change in stock status (kobe plot): red area in 2024 to green area in 2025
6. Catch quotas with different HCRs: 950-1300 kt.
7. MSY ~ 1000-1300 thousand tones
8. 2025 catch was 1296 thousand tones

Thanks to

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Dr. Gang Li for providing the updated of preliminary Chinese data

References

ICES 2021. Benchmark Workshop on the development of MSY advice for category 3 stocks using Surplus Production Model in Continuous Time; SPiCT (WKMSYSYSPICT). ICES Scientific Reports. Report. https://doi.org/10.17895/ices.pub.7919

Ichii, T., K. Mahapatra, T. Watanabe, A. Yatsu, D. Inagake e Y. Okada. 2002. Ocurrence of jumbo flying squid *Dosidicus gigas* aggregations associated with the countercurrent ridge off the Costa Rica Dome during 1997 El Niño and 1999 La Niña. Mar. Ecol. Prog. Ser. 231:151-166.

IMARPE 2022. Situación del calamar gigante durante el 2021 y perspectivas de pesca para el 2022. (Jumbo squid situation during 2020 and fishing prospects for 2021). IMARPE, Perú. Report in Spanish. https://www.gob.pe/institucion/imarpe/informes-publicaciones/3250952-situacion-del-calamar-gigante-durante-el-2021-y-perspectivas-de-pesca-para-el-2022

Kristensen, K., A. Nielsen, C. W. Berg, H. Skaug, B. M. Bell. 2016. TMB: Automatic Differentiation and Laplace Approximation. Journal of Statistical Software, 70(5), 1-21. doi:10.18637/jss.v070.i05

Mildenberger T.K., A. Kokkalis, C.W. Berg. 2021. Guidelines for the stochastic production model in continuous time (SPiCT). 07 September, 2021.

Payá I. 2022. Estatus y posibilidades de explotación biológicamente sustentables de los principales recursos pesqueros nacionales, año 2022: Jibia. (Status and possibilities of biologically sustainable exploitation of the main national fishery resources, year 2022: *D. gigas)*. Instituto de Fomento Pesquero. Chile. 101 pages and 4 annexes. In Spanish with abstract in English. https://doi.org/10.13140/RG.2.2.18924.85126

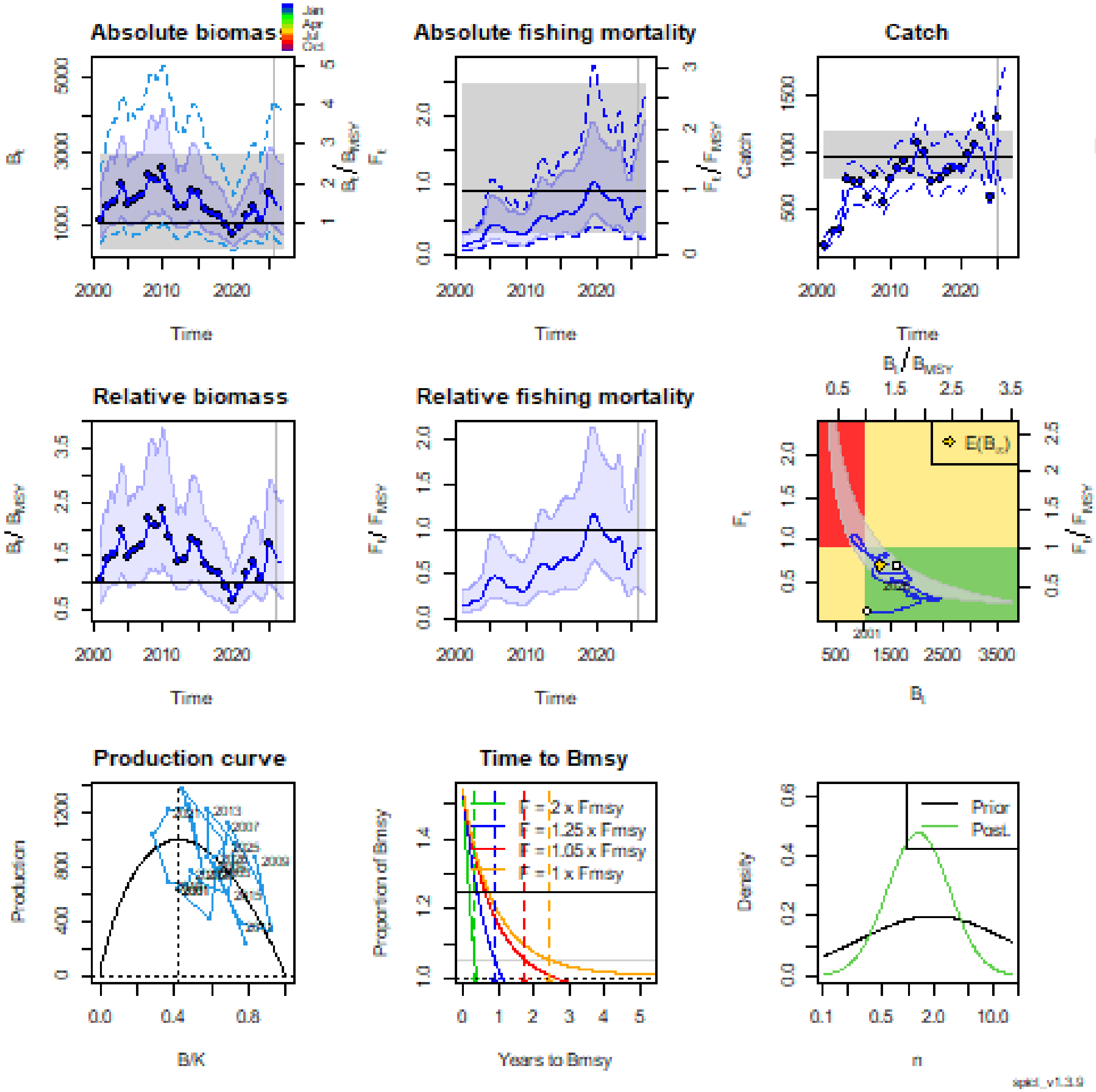
Payá I. 2022. First attempt to apply the Stochastic Production model in Continuous Time (SPiCT) to Dosidicus gigas Chile in FAO area 87. SC10. SPRFMO. www.sprfmo.int/assets/02-SC10/Meeting-Papers/SC10-SQ10-SPiCT-assessment-for-D-gigas-in-FAO-area-87.pdf.

Payá I. 2023. Estatus y posibilidades de explotación biológicamente sustentables de los principales recursos pesqueros nacionales, año 2023: Jibia. (Stock status and TAC 2023 for Chilean squid *Dosidicus gigas*). Segundo Informe Técnico Jibia 2023. Instituto de Fomento Pesquero. Chile. 101 pages and 4 annexes. In Spanish with abstract in English. http://dx.doi.org/10.13140/RG.2.2.29978.82885

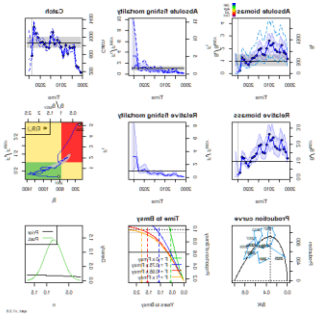
Pedersen M.W. and C.W. Berg. 2017. A stochastic surplus production model in continuous time. Fish and Fisheries 18 (2): 226–43. https://doi.org/10.1111/faf.12174.

Supplementary information

Case 1



Absolute biomass
Absolute fishing mortality
Catch
Relative biomass
Relative fishing mortality
Production curve
Time to Bmsy
F = 2 x Fmsy
F = 1.25 x Fmsy
F = 1.05 x Fmsy
F = 1 x Fmsy
Prior
Post.
Time
B/K
Years to Bmsy
Production
Proportion of Bmsy
Density
Catch
spict_v1.3.9



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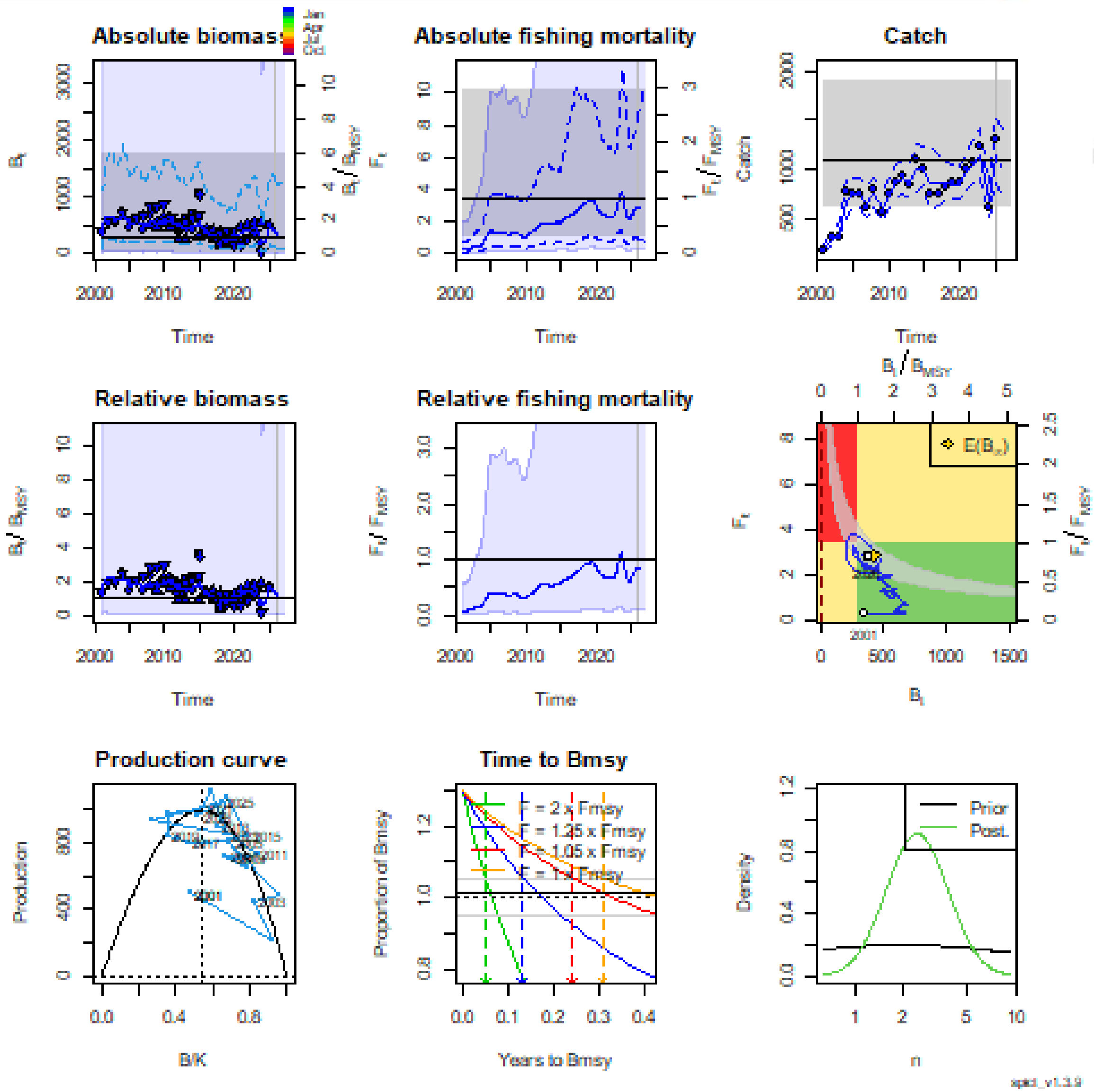
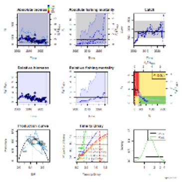
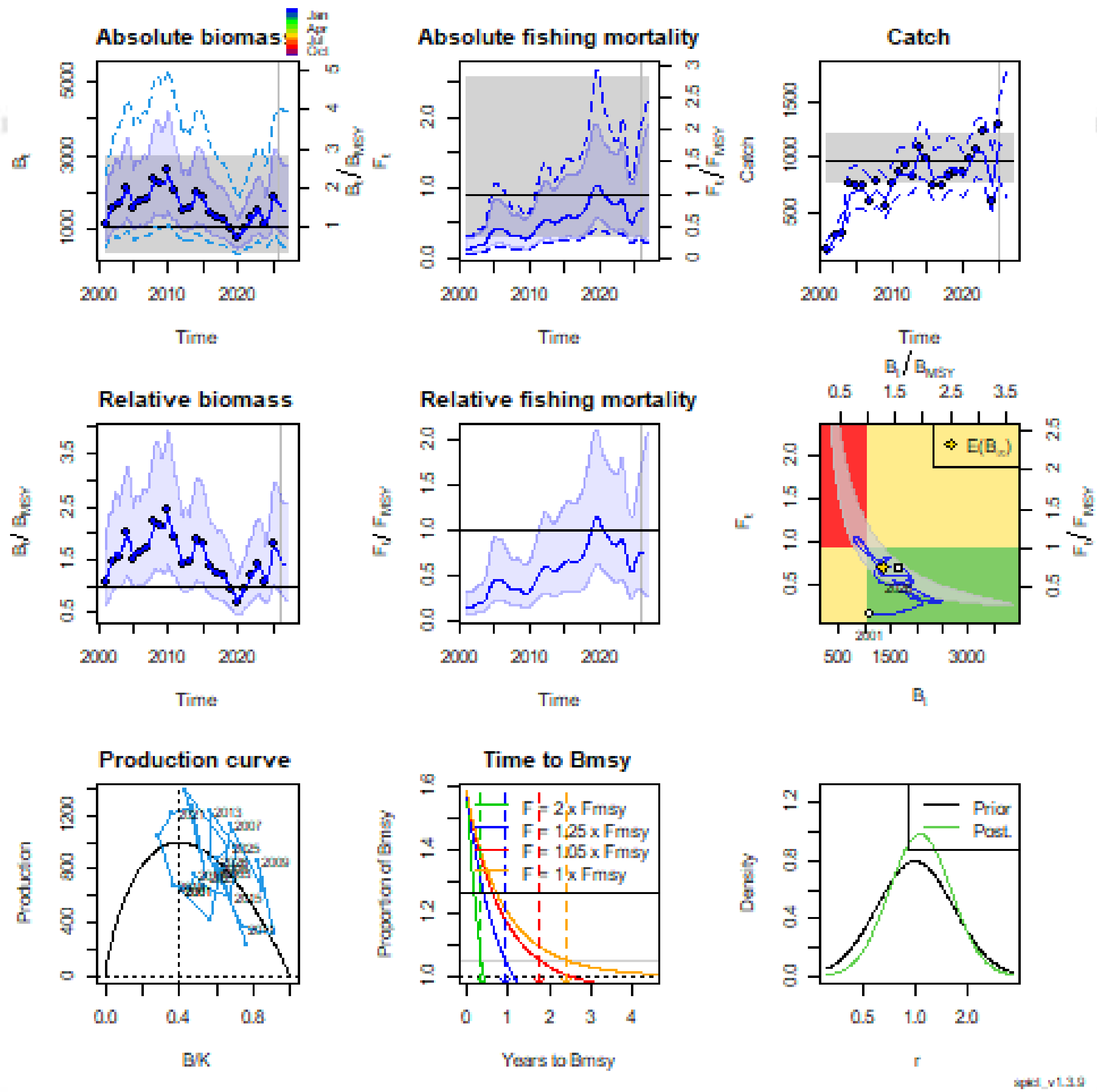


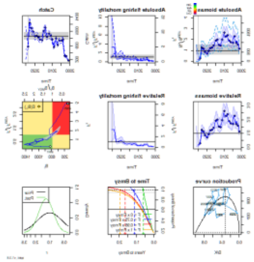
Absolute biomass
Absolute fishing mortality
Catch
Relative biomass
Relative fishing mortality
Production curve
Time to Bmsy
Time
B/K
Years to Bmsy
Production
Proportion of Bmsy
Density
Prior
Post.
F = 2 x Fmsy
F = 1.25 x Fmsy
F = 1.05 x Fmsy
F = 1 x Fmsy
spict_v1.3.9



Case 3



Absolute biomass
Jan
Apr
Jul
Oct
Absolute fishing mortality
Catch
Relative biomass
Relative fishing mortality
E(B∞)
Production curve
Time to Bmsy
F = 2 x Fmsy
F = 1.25 x Fmsy
F = 1.05 x Fmsy
F = 1 x Fmsy
Prior
Post.
Time
B/K
Years to Bmsy
Production
Proportion of Bmsy
Density
r
Catch
spict_v1.3.9



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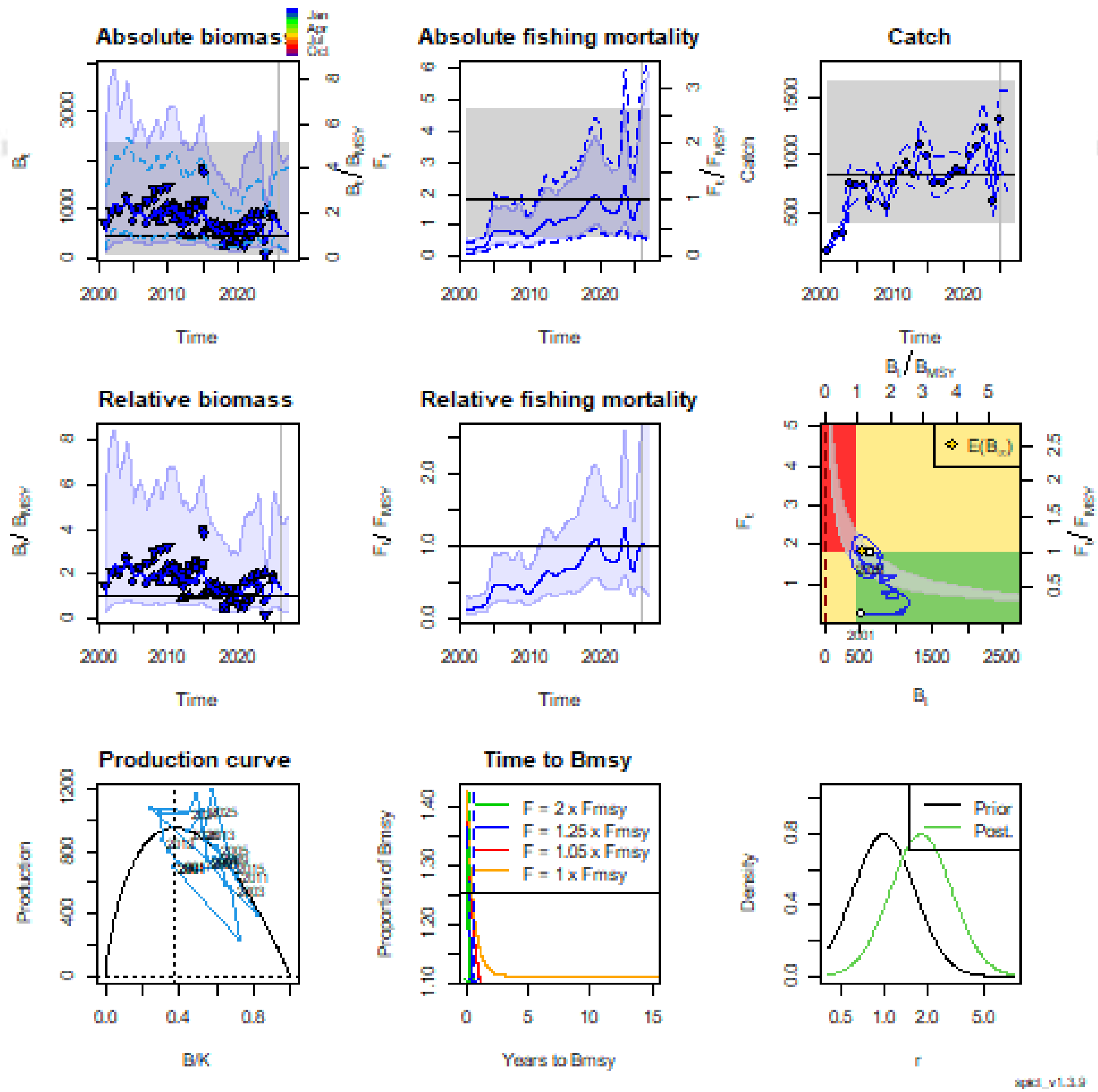
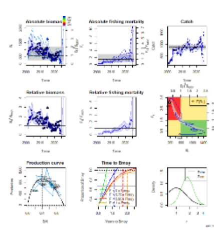


Absolute biomass
Jan
Apr
Jul
Oct
Absolute fishing mortality
Catch
Time
Relative biomass
Relative fishing mortality
E(B∞)
2001
Production curve
Production
B/K
Time to Bmsy
F = 2 x Fmsy
F = 1.25 x Fmsy
F = 1.05 x Fmsy
F = 1 x Fmsy
Proportion of Bmsy
Years to Bmsy
Prior
Post.
Density
r
spict_v1.3.9



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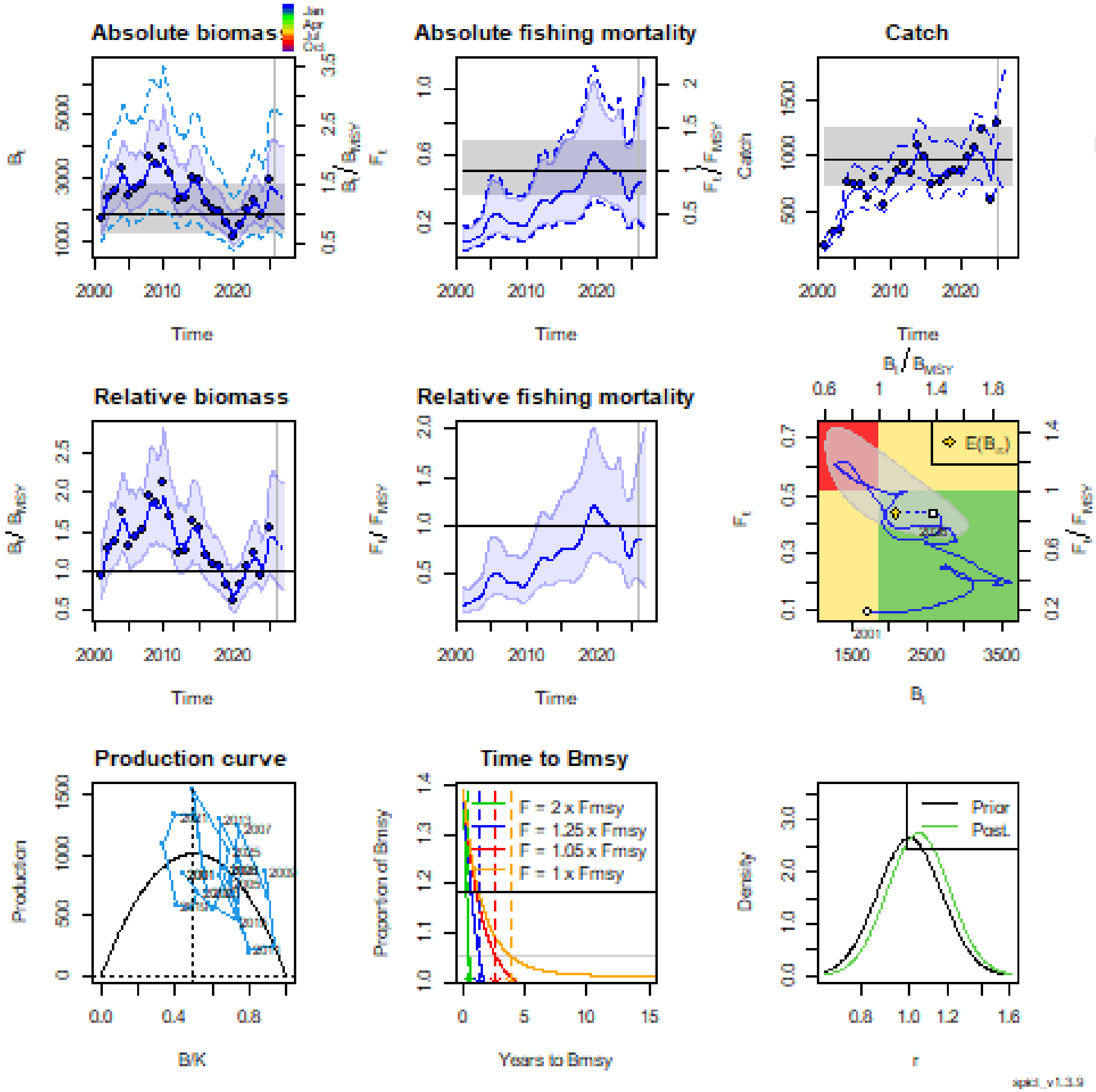
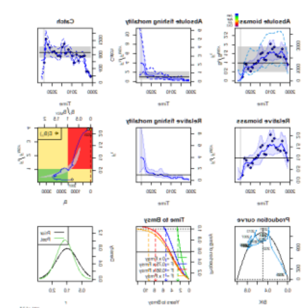
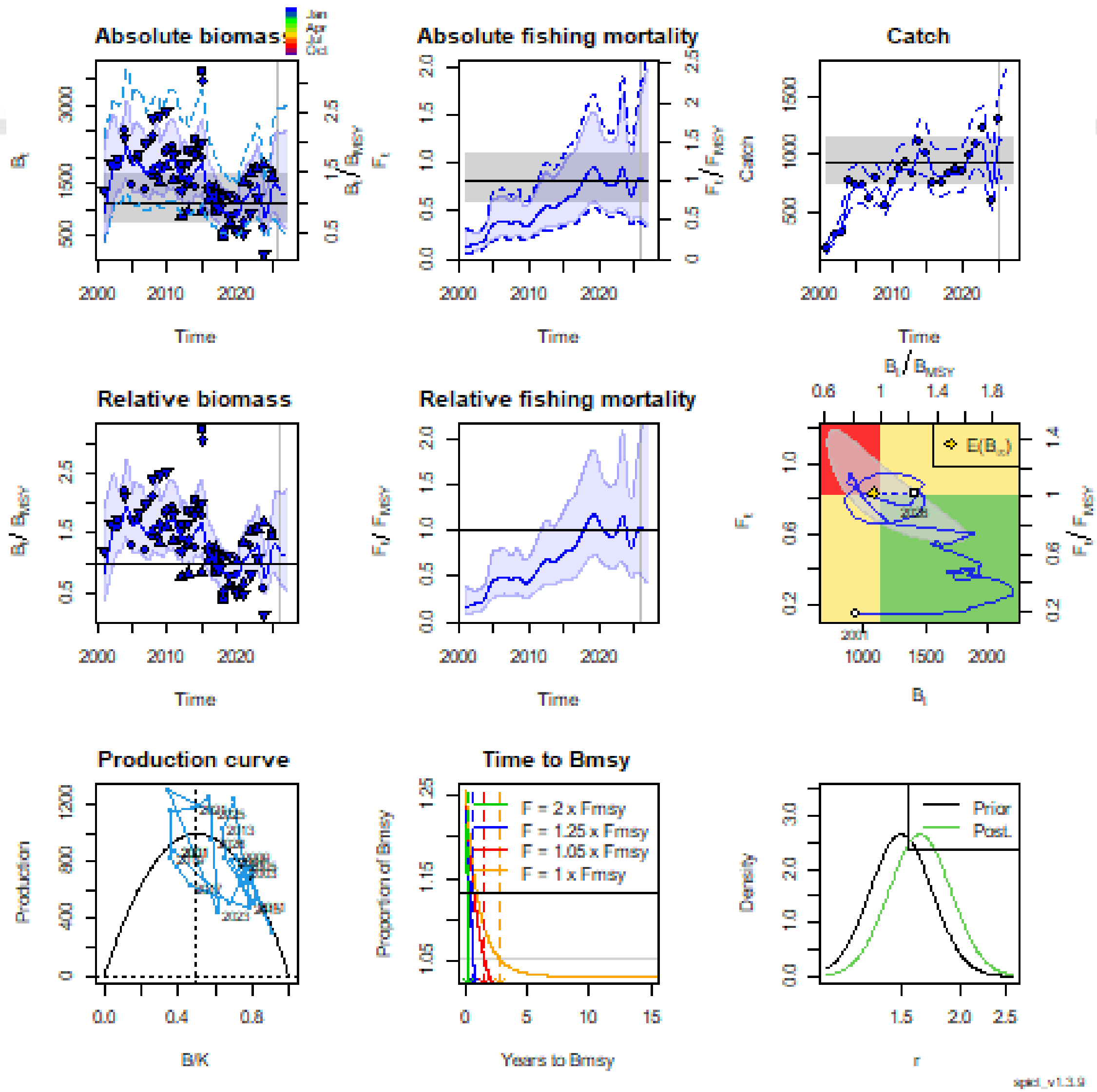
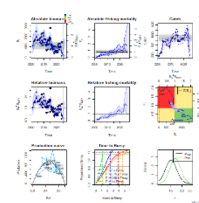

Absolute biomass
Jan
Apr
Jul
Oct
Time
Absolute fishing mortality
Time
Catch
Time
Relative biomass
Time
Relative fishing mortality
Time
E(B∞)
2001
Production curve
B/K
Time to Bmsy
F = 2 x Fmsy
F = 1.25 x Fmsy
F = 1.05 x Fmsy
F = 1 x Fmsy
Proportion of Bmsy
Years to Bmsy
Prior
Post.
Density
r
spict_v1.3.9

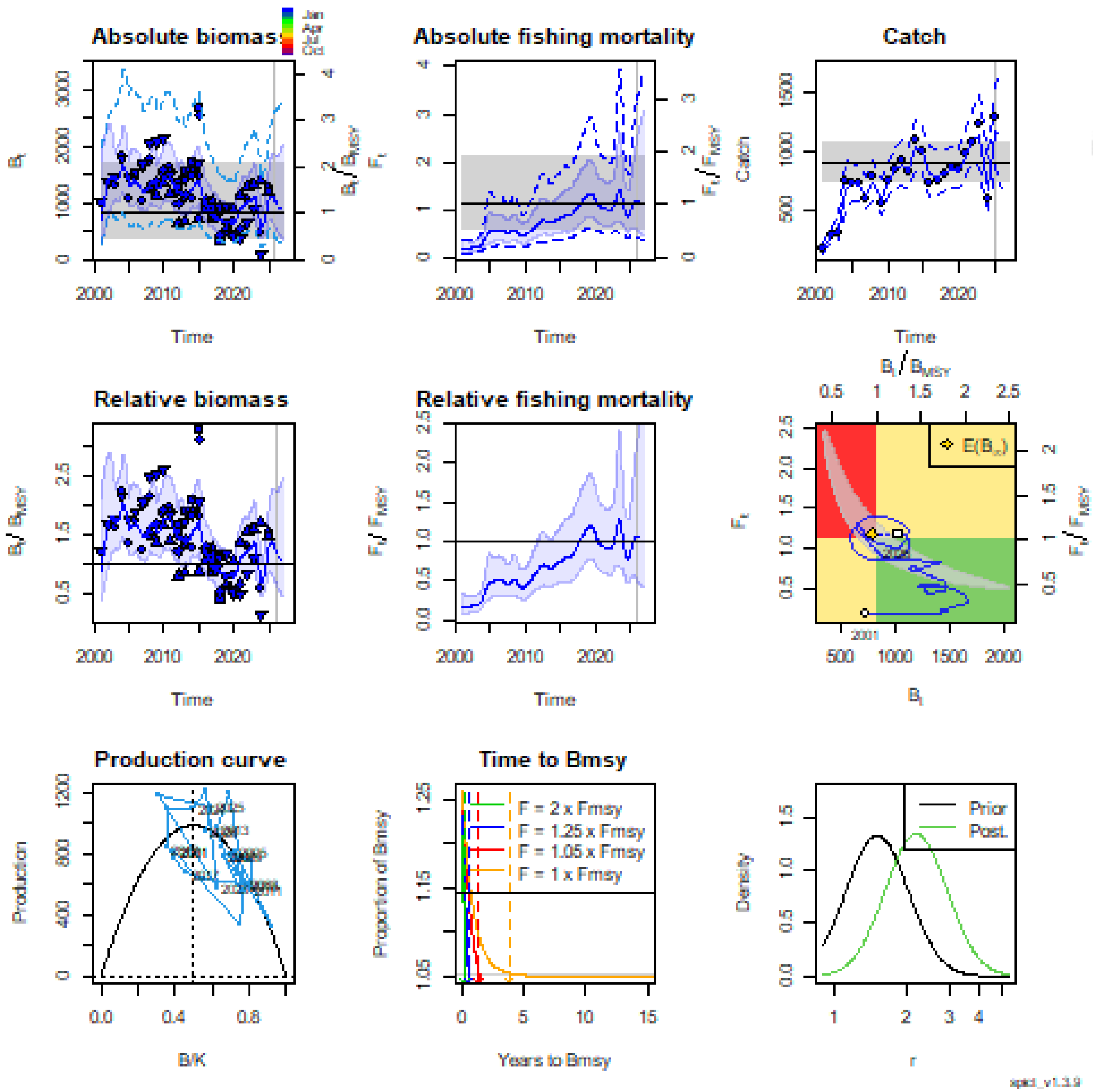


Case 6

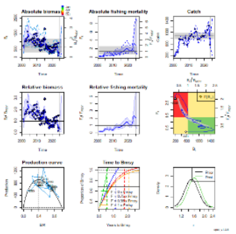

Absolute biomass
Jan
Apr
Jul
Oct
Absolute fishing mortality
Catch
Relative biomass
Relative fishing mortality
E(B∞)
Production curve
Time to Bmsy
F = 2 x Fmsy
F = 1.25 x Fmsy
F = 1.05 x Fmsy
F = 1 x Fmsy
Prior
Post.
Time
Years to Bmsy
Proportion of Bmsy
Production
Density
Catch
B/K
spict_v1.3.9



Case 7



Absolute biomass
Jan
Apr
Jul
Oct
Absolute fishing mortality
Catch
Relative biomass
Relative fishing mortality
$E(B_\infty)$
Production curve
Time to Bmsy
F = 2 x Fmsy
F = 1.25 x Fmsy
F = 1.05 x Fmsy
F = 1 x Fmsy
Prior
Post.
Time
B_t
B_t / B_{MSY}
F_t
F_t / F_{MSY}
Catch
Production
B/K
Proportion of Bmsy
Years to Bmsy
Density
r
2001
spict_v1.3.9



r and K by case

r parameter by case.

	estimate	cilow	ciupp	log.est
1	1.2	0.35	4.18	0.18
2	6.32	1.86	21.5	1.84
3	1.07	0.48	2.36	0.07
4	1.82	0.68	4.87	0.6
5	1.04	0.79	1.39	0.04
6	1.65	1.23	2.21	0.5
7	2.18	1.22	3.9	0.78

K (kt) parameter by case.

	estimate	cilow	ciupp	log.est
1	2632	1229	5633	8
2	716	270	1903	7
3	2755	1435	5289	8
4	1447	749	2799	7
5	3879	2611	5765	8
6	2420	1659	3530	8
7	1810	967	3385	8

Bmsy and Fmsy by case

Bmsyd (kt) by case.

	estimate	cilow	ciupp	log.est
1	1109	402	3064	7
2	386	145	1028	6
3	1096	386	3110	7
4	544	245	1207	6
5	1940	1305	2882	8
6	1210	829	1765	7
7	905	484	1692	7

Fmsyd (kt) by case.

	estimate	cilow	ciupp	log.est
1	0.9	0.34	2.43	-0.1
2	2.59	0.99	6.77	0.95
3	0.91	0.32	2.55	-0.09
4	1.75	0.8	3.81	0.56
5	0.52	0.39	0.69	-0.65
6	0.82	0.61	1.1	-0.2
7	1.09	0.61	1.95	0.09

MSY by case

MSYd (kt) by case.

	estimate	cilow	ciupp	log.est
1	1001	805	1244	7
2	999	875	1140	7
3	998	801	1243	7
4	951	823	1099	7
5	1012	764	1340	7
6	995	794	1248	7
7	987	823	1183	7